WILL

Relational Geometry

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Abstract

This paper, the first in the **WILL RG Trilogy:**
Part I: Relational Geometry;
Part II: Relational Cosmology;
Part III: Relational Quantum Mechanics (R.Q.M.).
By applying extreme methodological constraints we establish *Relational Geometry (RG)*: a background-free foundation and its algebraic formulation **Relational Orbital Mechanics** that reproduces the 1PN post-Newtonian expansions of General Relativity — including perihelion precession, light deflection, and frame-dragging — without the use of metric tensors, Christoffel symbols, or numerical geodesic integration, and produces a specific $\mathcal{O}(\beta^4)$ next order term that differs from 2PN GR and provides a falsifiable signature for future high-precision observations. RG is developed here not as a competitor to General Relativity but as a step in its natural evolution: an attempt to complete the background-independent program that Einstein himself aimed for. From first principles we reproduce Lorentz factors, the energy-momentum relation, Minkowski and Schwarzschild metric intervals via the dimensionless projections of kinematic and potential state difference. All known GR critical surfaces (photon sphere, ISCO, horizons) emerge as simple fractions of these projections.

***This paper* offers new potential solutions to several long-standing problems, including:**

- **Derivation of Relational Orbital Mechanics** (Background-free and algebraically closed),
- **Resolution of gravitational singularities** (via bounded radial distance $r_{min} = R_s/2$ (5.9)),
- **Derivation of the equivalence principle** (from the rest-invariant state scaling, 4.2),
- **Removal of local energy density ambiguity** $\rho = \frac{\kappa^2 c^2}{8\pi G r^2}$ (5.7.1),
- **Revelation of a clear relational symmetry between kinematic and potential** energies (3.9),
- **Establishment of a computationally simpler and ontologically consistent foundation** (for subsequent papers on cosmology (Part II) and quantum mechanics (Part III)).

A Direct Note to the Reader: Why engage with this research?

Engaging with unknown independent research is a substantial commitment, and you should not do it blindly. **I recommend the following:** Define the criteria this research must satisfy to justify the investment of your time. Then, ask WILL-AI if this work meets your standards. The AI is trained on these papers and works as an advanced search engine. It will give you an honest answer within its limitations, and will provide direct links to related sections, Colab notebooks and Desmos projects. If you are still in doubt, browse **willrg.com**:

- **Falsifiable Predictions** - from atomic structure through the local neighbourhood to the cosmic horizon;
- **Galactic Dynamics** - test rotation curves predictions against **SPARC dataset**;
- **Holographic Decoder** - **Breaking** $M \sin(i)$ **Degeneracy** with 1D RV data string;
- **Computational Efficiency** - side-by-side comparison with General Relativity;
- **Logos Map** - trace the derivation logical chain;
- **WILL-AI** - just have a chat — it's fun.

Regardless of your decision, I wish you to find what you are seeking.

Anton Rize

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Einstein on Emergence of Spacetime

"There is no such thing as an empty space, i.e., a space without field. . . . Space-time does not claim existence on its own, but only as a structural quality of the field."
— *Albert Einstein*

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Derivation Chain Map

Methodological constraints:

↓(*Epistemic Hygiene+Relational Origin+Ontological Minimalism+Math Transparency*)↓

Relational Closure + Causal Continuity + Isotropy Theorems

↓ *(apply to existing physics)* ↓

Standard Primitives: $\underbrace{\text{background manifold + metric}}_{\text{spacetime structure (not derived)}} \quad + \quad \underbrace{\text{G, c, M, fields}}_{\text{energy dynamics (postulated)}}$

↓ *(Ontological reduction by Relational Origin Principle — no background allowed)* ↓

One Primitive: ***WILL*** ≡ ***SPACE-TIME-ENERGY*** $\Longrightarrow$ $\boxed{\text{SPACETIME} \equiv \text{ENERGY}}$

↓ *(by Relational Closure + Causal Continuity + Isotropy: derive relational energy carriers)* ↓

S^1 **(1-DOF kinematic) +** S^2 **(2-DOF potential)**

↓ *(by Duality of Relation Lemma)* ↓

Relational Conservation Theorem: $\underbrace{\text{Amplitude}^2}_{\text{External Transformation}} + \underbrace{\text{Phase}^2}_{\text{Internal Change}} = 1$

$\boxed{\underbrace{1/\beta_Y = 1 + z_\beta}_{\text{transverse Doppler shift}} \quad \underbrace{\beta^2 + \beta_Y^2 = 1}_{\text{Kinematic } S^1}}$ **Relational Projections** $\boxed{\underbrace{\kappa^2 + \kappa_X^2 = 1}_{\text{Potential } S^2} \quad \underbrace{1/\kappa_X = 1 + z_\kappa}_{\text{gravitational redshift}}}$

↓ *(by DOF-Indifference Lemma: equal quadratic weight to each independent DOF)* ↓

Closure Theorem: $\underbrace{1\,\text{Amplitude}^2}_{\text{on } S^2} = \underbrace{2\,\text{Amplitude}^2}_{\text{on } S^1} \Longrightarrow \boxed{\kappa^2 = 2\beta^2}$

↓ *(by Relational Conservation Theorem derive metric intervals as scaffolding)* ↓

Minkowski: $\underbrace{\beta^2 + \beta_Y^2 = 1}_{\text{irreducible primitive}} + \underbrace{x, y, z, t...}_{\text{coordinate inflation}} = \underbrace{c^2 d\tau^2 = c^2 dt^2 - dx^2 - dy^2 - dz^2}_{\text{mathematical scaffolding}}$

Schwarzschild: $\underbrace{\kappa^2 + \kappa_X^2 = 1}_{\text{irreducible primitive}} + \underbrace{x, y, z, t...}_{\text{coordinate inflation}} = \underbrace{c^2 d\tau^2 = c^2(1 - (R_s/r))\, dt^2}_{\text{mathematical scaffolding}}$

↓ *(by Relational Closure and Causal Continuity Theorems: any change must be balanced)* ↓

Energy Symmetry Law: $\boxed{\Delta E_{A\to B} + \Delta E_{B\to A} = 0}$

$\downarrow \Delta E_{A\to B} = E_0(\kappa_{X,B}/\beta_{Y,B} - \kappa_{X,A}/\beta_{Y,A}); \quad \Delta E_{B\to A} = E_0(\kappa_{X,A}/\beta_{Y,A} - \kappa_{X,B}/\beta_{Y,B}) \downarrow$

Speed of light: $\beta \to 1 \;\to\; \beta_Y \to 0 \;\to\; E = E_0 \cdot \kappa_X/0 \to \infty \;\to$ **requires** ∞ **energy**

Event Horizon: $\kappa \to 1 \;\to\; \kappa_X \to 0 \;\to\; E = E_0 \cdot 0/\beta_Y \to 0 \;\to$ **requires** 0 **energy**

↓ *(by Unified Relational Scaling Lemma)* ↓

Equivalence Principle $\boxed{m_g \equiv m_i \equiv m = E_0/c^2}$

↓ *(**Lagrangian and Hamiltonian** derived as collapsed single-point approximations)* ↓

$\underbrace{\Delta E_{A\to B} + \Delta E_{B\to A} = 0}_{\text{exact symmetry law}} \approx \underbrace{\tfrac{1}{2}(\kappa_A^2 - \kappa_B^2) + \tfrac{1}{2}(\beta_B^2 - \beta_A^2)}_{\text{first order approximation}} \approx \underbrace{T \pm V}_{\text{ontological collapse}}$

↓ *(apply RG to orbital mechanics)* ↓

Relational Orbital Mechanics $\boxed{\textbf{R.O.M.}}$**: a closed algebraic system of equations**

↓*(by Factorization Theorem: every quantity X splits into unitless core $\hat{X}$ and scale label S_X)*↓

Unitless Core: $\underbrace{X = \hat{X}(,e)}_{\text{unitless core}} \underbrace{S_X(R_s, c, G)}_{\text{units-scale}}$ **G and M are unit-conversion factors.**

↓ *(by Inverse Square, Closure and Causal Continuity Theorems)* ↓

No Singularities: $\beta_{\max}^2 = 1 \;\to\; \kappa_{\max}^2 = 2 \;\to\; r_{\min} = R_s/\kappa_{\max}^2 \;\to\; r_{\min} = R_s/2$

The point $r = 0$ **is absent from the admissible relational range**

↓ *(apply 2D to 3D translation interfaces to derive Relational Field Equation:)* ↓

$\boxed{\kappa^2 = R_s/r = \rho_{\text{field}}/\rho_{\max}} \Longrightarrow \boxed{\text{Spacetime Geometry}(R_s/r)} \equiv \boxed{\text{Energy Density}(\rho/\rho_{\max})}$

↓ **Theoretical Ouroboros** ↓

$\boxed{\text{SPACETIME} \equiv \text{ENERGY}} \Leftrightarrow \boxed{\text{SPACETIME GEOMETRY} \equiv \text{ENERGY DENSITY}}$

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This document must be read **literally**.

- All terms are defined within **Relational Logic:** *Positions, velocities and other properties of objects are only meaningful relative to other objects. Space without objects is operationally undetectable and physically meaningless, the same way time is meaningless without events. Within relational logic space and time are systems of relations that exist between objects.*
- Any attempt to reinterpret these derivations through conventional substantival metric notions (*absolute distances, external backgrounds, hidden containers*) will produce distortions and mathematical misreadings.
- **Take the words as written** and ontology as derived, without importing external concepts.

Remark 0.1 (**The term "geometry"**)**.** *Is used herein in its fundamental, classical sense: the pure logic of relations, ratios, and structural constructions. It* **does not** *refer to modern differential geometry. There is no* a priori *metric tensor ($g_{\mu\nu}$), no Riemannian manifold, and no underlying coordinate grid. The geometry emerges from physical relations, not from a background space.*

1 Stage 0: The Last Geocentric Epicycle

General Relativity remains the most successful description of gravitation we possess, and its author regarded the elimination of background structure as central to its meaning: Einstein insisted that space-time "does not claim existence on its own, but only as a structural quality of the field." The analysis below is therefore aimed not at General Relativity itself, but at the scaffolding it inherited. What follows is offered as the next step in GR's own evolution — an attempt to complete the relational program Einstein began.

The standard formulation of General Relativity relies on the concept of a pre-existing, container-like spacetime that then gets "filled" with fields and matter.

Derivation of the Einstein field equations begins with the Einstein-Hilbert action, which is built upon the metric as the fundamental variable. This metric is defined on a smooth manifold that is assumed to exist *a priori*. This manifold, even without a metric, carries topological and differential structure - an absolute scaffold. This constitutes surplus ontology because it introduces an entity (the manifold) that is neither derived nor observed.

The stress-energy tensor is derived from the variation of the matter Lagrangian with respect to the metric. This assumes that energy is a property of matter fields that can be localized in spacetime. However, this localization is frame-dependent (via the equivalence principle) and leads to well-known problems such as the non-uniqueness of the gravitational energy-momentum pseudotensor. Energy is treated as an absolute property of matter rather than a relational measure. This remains a metaphysical assumption at the heart of modern physics.

1.1 The contemporary split: an inherited ontological commitment

All present-day theories (SR, GR, QFT, ΛCDM, Standard Model) are built with a *bi-variable* syntax:

$$\underbrace{\text{fixed manifold} + \text{metric}}_{\text{structure}} + \underbrace{\text{fields} + \text{constants}}_{\text{dynamics}}.$$

No observation demands this duplication; it is retained purely because the resulting Lagrangians are empirically adequate *inside* the split. The split is therefore *not* an empirical discovery but an inherited, untested ontological assumption.

1.2 Empirical deficit of the separation

- **Local energy conservation** verified only *after* the metric is declared fixed; no experiment varies the *volume* of flat space and checks calorimetry.
- **Universality of free fall** tests $m_i = m_g$ numerically, not the claim that inertia resides *in* the object rather than in a geometric scaling relation.
- **Gravitational-wave polarisations** test spin content, not ontology; extra modes can still be called "matter on spacetime".
- **Casimir/Lamb shifts** measure *differences* of vacuum energy between two geometries; the absolute bulk term is explicitly subtracted, leaving the split intact.

Each of these tests is an *internal consistency check* of a formalism that already presupposes two substances. None constitutes *positive evidence* for the split.

1.3 Historical Pattern: breakthroughs delete, not add

- **Copernicus** eliminated the Earth/cosmos separation.
- **Newton** eliminated the terrestrial/celestial law separation.
- **Maxwell** eliminated the electricity/magnetism separation.
- **Einstein** eliminated the space/time separation.

Each step widened the relational circle and reduced the number of primitives — the unexplained absolutes. The spacetime–energy split is the only survivor of this pruning sequence.

1.4 Consequence

Until an experiment varies the amount of space while holding everything else fixed, the spacetime–energy separation remains an *untested metaphysical postulate*. **It is the last geocentric epicycle in physics.** Removing it is not a rejection of General Relativity; it is the same pruning operation that produced General Relativity in the first place.

2 Stage I: Ontological Construction (The Primitives)

> *Every logical step is subject to empirical audit: if any consequence of this chain contradicts observation, the chain fails.*

As humans, we often project ourselves onto observed phenomena, obscuring the true nature of reality behind our anthropic tendencies. Theoretical development requires defined epistemic constraints to prevent the introduction of unobservable parameters and anthropocentric interpretations.

While traditional methods introduce new fields, particles, or a priori postulates to resolve discrepancies with empirical data, this research relies exclusively on operational realism. Any absolute structure lacking direct empirical observability is refused.

Primary Goal: Predict empirically observed physical reality exclusively from the relational necessities that remain after unproven absolute structures removed.

We derive predictions directly from scientific reasoning, demonstrating that unobservable mathematical artifacts are redundant. The principles that follow are not assumed axioms; they are the deductive consequence of enforcing the burden of proof. Empirical alignment remains the sole arbiter of this methodology.

2.1 Foundational Methodological Principles

Flow chart of derivation logic
Logos Map

Principle 2.1 (Epistemic Hygiene)**.** Epistemic Hygiene as Refusal to Import Unjustified Assumptions. *This line of reasoning derives physics by* ***removing hidden assumptions****, rather than introducing new postulates. This construction is deliberate and contains zero free parameters. This is not a simplification - it is a deliberate epistemic constraint. No assumptions are introduced and no constructs are retained unless they are geometrically or energetically necessary.*

Principle 2.2 (Relational Origin)**.** All physical quantities must be defined by their relations. *Any introduction of absolute properties risks reintroducing metaphysical artefacts and contradicts the foundational insight of relationalism.*

Principle 2.3 (Ontological Minimalism)**.** *Maintain the minimum number of ontological assumptions and foundational primitives. Absolute backgrounds — spatial container, axis of time, absolute distance, forces, fields — are excluded as preexisting primitives but may appear as translation into common descriptive vocabulary.*

Principle 2.4 (Mathematical Transparency)**.** *Every mathematical phrase, operation, or identity is treated as a binding ontological statement, locking the formalism to always reflect direct physical meaning preventing obscure mathematical constructs. Every symbol must be anchored to unique physical idea with transparent ontological origin. Introducing symbols without explicit necessity constitutes* ***Over-parameterization****: the proliferation of symbols without corresponding physical meaning. Number of symbols = Number of independent physical ideas.*

> *Mathematics is a language, not a world. Its symbols must never outnumber the physical meanings they encode.*

A Note on Methodology vs. Ontology

It may be objected that the foundational principles constitute axioms, thereby contradicting the claim of *zero assumptions*. This objection conflates *methodological constraints* with *ontological axioms*.
Under the strictures of the Scientific Method, the burden of proof rests exclusively on the claimant asserting an absolute existence. Refusing to import unobserved absolutes (such as a background container or intrinsic properties) is not an assumption; it is the default epistemic stance — the null hypothesis.
These principles function strictly as rules of logical engagement to prevent the introduction of surplus ontology. To challenge them is to argue for the admission of a specific unobserved absolute — and under the empirical foundation of physics, the burden of proof rests with that admission.

2.1.1 The Foundational Core: Relational Geometry

These foundational methodological principles lead us to the core of Relational Geometry:

Theorem 2.5 (Relational Closure). *A purely relational system cannot be geometrically open; any stable, self-contained system must be described by a closed and self-consistent relational geometry. Its geometry is closed by absolute analytic necessity.*

Proof by Analytic Tautology.

1. **Premise 1 (Relational Origin):** Within RG Physical quantities are defined exclusively by their relations (Principle 2.2).
2. **Premise 2 (Interaction as Integration):** If a hypothetical "outside" entity or boundary were to physically interact with the system, that interaction itself constitutes a relation.
3. **Deduction (The Tautology):** As relation is formed, it is mathematically and physically integrated into the system.

An "open relational system" experiencing external influence is a logical contradiction. The system is closed not by a spatial boundary or a metric container, but by the strict impossibility of an interacting non-relation. The system is relationally closed by definition. □

Theorem 2.6 (Causal Continuity (Conservation)). *Within any closed dynamical system there must exist an invariant quantitative measure of change. Without an external background any change must be perfectly balanced by a complementary change elsewhere in the system enforcing* ***Conservation and Causal Continuity****.*

Proof by Structural Necessity.

1. **Premise 1 (Relational Origin):** Physical quantities are defined exclusively by their relations (Principle 2.2).
2. **Premise 2 (Empirical Change):** States undergo observable change.
3. **Premise 3 (Relational Closure):** As proven in Theorem 2.5, the relational system is analytically closed. There is no external background, void, or hidden container to absorb or inject influence.
4. **Deduction (Causal Continuity):** Because the system is closed (Premise 3), an empirical change (Premise 2) cannot vanish into a void or emerge from one. To maintain the structural integrity of the relations (Premise 1), any change must be perfectly balanced by a complementary change elsewhere in the system.

Therefore, within this closed relational network, there must exist an invariant quantitative measure of change. □

2.1.2 What is Energy in Relational Ontology?

Across all domains of physics, one empirical fact persists: in every closed system there exists a quantity that never disappears or arises spontaneously, but only transforms in form. This invariant is observed under many guises — kinetic, potential, thermal, chemical — yet all are interchangeable, pointing to a single underlying structure.

Crucially, this quantity is never observed directly, but only through *differences between states*: a change of velocity, a shift in configuration, a transition of phase. Its value is relational, not absolute: it depends on the chosen frame or comparison, never on an object in isolation.

Moreover, this quantity provides continuity of causality. If it changes in one part of the system, a complementary change must occur elsewhere, ensuring the unbroken chain of transitions. Thus it is the bookkeeping of causality itself — the ledger of change.

Corollary 2.7 (Energy). *In classical and relativistic mechanics, this derived invariant measure of change is assigned the historical label "Energy". By showing its relational origin, we strip "Energy" of its substantialist properties (as a "substance" or "thing") and reveal its origin as the necessary measure of change and bookkeeping ledger of causality.*

Energy :

Definition 2.8 (Energy).

$$\boxed{\begin{array}{c}\textbf{\textit{Energy is the invariant relational measure}}\\ \textbf{\textit{of state transformation}}\\ \textbf{\textit{within closed system.}}\end{array}}$$

Remark 2.9 (Energy Ledger: intuitive understanding). *Consider a completely closed economy with no physical currency — no dollar bills, no gold coins, and no central bank (acting as an absolute container). It is a purely relational economy consisting solely of a massive ledger of debts and credits between individuals. If one account decreases, another must increase. If the system is frozen at any given moment, the sum of all debts and credits across the entire ledger must equal zero. "Money" does not exist as a physical object; it is simply the mathematical rule that the ledger must always balance. Energy functions identically: it is not a substance, but the invariant rule of relational accounting.*

Theorem 2.10 (Relational Isotropy). *Within a background-free relational system, no spatial direction or coordinate can be* a priori *privileged. Therefore, the geometry encoding the relational balance (energy) must be maximally symmetric (isotropic).*

Proof by Relational Origin.

1. **Premise 1 (Absence of a Grid):** As established by Theorem 2.5 (Relational Closure), the relational system possesses no external container, absolute metric tensor, or underlying coordinate grid.
2. **Premise 2 (Direction requires Relation):** Under the Principle of Relational Origin (Principle 2.2), "direction" or "orientation" cannot exist in a void; it only physically manifests as a specific interaction or measurement between defined participants.
3. **Deduction (Pure Gauge):** Because there is no absolute grid (Premise 1), assigning an intrinsic, absolute axis to the unmeasured relational ledger (energy) introduces unobservable "surplus ontology." Any such arbitrary orientation is a mathematical artifact (pure gauge).

Therefore, prior to a specific interaction defining a local axis, the pure relational structure encoding this conserved energy ledger must possess no intrinsic privileged direction. It must be strictly isotropic. □

2.2 Unifying Ontological Principle

Summary

Any attempt to treat *"spacetime structure"* as separate from *"dynamics"* quietly imports a background container that is not justified by the phenomena. This violates epistemic hygiene: it introduces an ontological artifact without necessity. Eliminating this separation compels the identification of structure and dynamics as two aspects of a single entity.

Lemma 2.11 (False Separation). *Any model that treats processes as unfolding* within *an independent background necessarily assigns to that background structural features (metric, orientation, or frame) not derivable from the relations among the processes themselves. Such a background constitutes an* extraneous absolute.

Proof. Suppose an independent background exists. Then at least one of its structural attributes - metric relations, a preferred orientation, or a class of inertial frames - remains fixed regardless of inter-process data. This attribute is not relationally inferred but posited a priori. It thereby violates the relational closure principle: it introduces a non-relational absolute external to the system. Hence the separation is unwarranted. □

Corollary 2.12 (Structure–Dynamics Coincidence). *Without introducing new primitives 2.11, we should see structure and dynamics as a single ontological element. The structural arena and the dynamical content must be identified: structure is dynamics, energy is spacetime.*

This leads us to a principle with *negative ontological weight*: it reduces the number of primitives and removes the separation between structure and dynamics. Spacetime is not a "container" but relational tension between energy states.

Principle 2.13 (Unifying Ontological Principle).

$$\boxed{\textbf{\textit{SPACETIME}} \equiv \textbf{\textit{ENERGY}}}$$

No background container permitted.

Definition 2.14 (WILL). **WILL** $\equiv$ **SPACE-TIME-ENERGY** *is the technical term we use for the unified relational structure determined by Unifying Principle 2.13. All physically meaningful quantities are relational features of WILL.*

The Last Geocentric Epicycle

Definition 2.15 (Ontological weight)**.** *The number of irreducible primitives a theory posits.*

Picture	Primitives
Substantival: container + content (spacetime, energy/matter)	≥ 2
Relational: WILL (one self-relating structure)	1

It is the operation physics has performed at every unification:

- **Heat** $\equiv$ molecular motion (deletes caloric);
- **Electricity** $\equiv$ magnetism (Maxwell);
- **Space** $\equiv$ time (Einstein).

Each merged two primitives into one. **SPACETIME** $\equiv$ **ENERGY** applies the identical operation. **This Principle does not add, it subtracts: it removes the hidden assumption. Structure and dynamics are two aspects of a single entity that we call WILL.**

Remark 2.16 (Auditability)**.** *Principle 2.13 is foundational but testable: it is subject to: (i) geometric audit (internal logical consequences) and (ii) empirical audit (agreement with empirical data).*

3 Stage II: Geometric Derivation

3.1 WILL Structure

Having established Principle 2.13 by removing the unwarranted separation of structure and dynamics, we are left with a single foundational primitive: ***WILL*** $\equiv$ ***SPACE-TIME-ENERGY***. Now we proceed to derive the necessary geometric and physical properties of ***WILL***.

The foundational core (2.1.1) — Closure, Causal Continuity, and Isotropy — establishes the necessary geometric properties for the relational energy carriers.

Relational Carrier Conventions:

All references to "carriers" within WILL RG open research are to be read in the strict relational sense:

- **Degree of freedom (DOF):** An n-dimensional relational carrier encoding the conserved transformation ledger.
- **Direction:** An oriented relation. Opposite orientations are physically distinct and not identified.
- **Closed carrier:** Compact and without boundary; relational ledger cannot leak into an external void.
- **No background:** No external embedding space. All geometric structure is reconstructible solely from relations between participants.
- **Maximal symmetry:** The carrier is homogeneous and isotropic with respect to oriented directions.
- **Minimal relational carrier:** A closed, maximally symmetric carrier utilizing exactly the required DOF without arbitrary parameters.

3.2 Deriving Relational Carriers

Theorem 3.1 (Minimal Relational Carriers of the Conserved Relational Ledger)**.** *The minimal relational carriers satisfying Closure, Conservation, Maximal Symmetry, and Mathematical Transparency are:*

(a) S^1 for directional (Kinematic) relational transformation;

(b) S^2 for omnidirectional (Potential) relational transformation.

Proof. The proof classifies the minimal types of relations and applies the derived geometric constraints:

The minimal non-trivial relation involves 1 degree of freedom (1-DOF): a sequence of state transformations.

1. 1-DOF (Kinematic) carrier must be:
   - **Closed** (by Relational Closure Theorem 2.5 — no boundary)
   - **Maximally symmetric** (by Isotropy Theorem 2.10 — no privileged point or direction on the carrier)

A linear segment cannot carry this relation because it possesses absolute endpoints. Endpoints act as boundaries where the transformational ledger would either vanish (violating the Conservation Theorem) or require an external bounding mechanism to reflect it (violating the Relational Origin and Ontological Minimalism principles 2.3, 2.2). **The unique 1-DOF boundaryless, isotropic topology is the circle (S^1).**

The next minimal relation involves 2 degrees of freedom (2-DOF): the omnidirectional distribution of the transformational ledger from a primary state.

2. **2-DOF (Potential) Relation carrier must be:**

   - **Closed** (by Relational Closure Theorem 2.5 — compact, no boundary, no leakage into an external void.)
   - **Maximally symmetric** (by Isotropy Theorem 2.10 — isotropic distribution of potential.)
   - **Orientable** (by Causal Continuity Theorem 2.6 — opposite directions are physically distinct, not identified)

   Topological classification of a boundaryless 2-carrier under the requirement of maximal symmetry eliminates all anisotropic surfaces (e.g., the torus T^2, which possesses distinct major and minor intrinsic axes). Orientability exclude Real Projective Plane ($\mathbb{RP}^2$) **The unique boundaryless, maximally symmetric 2-DOF topology is the 2-sphere (S^2).**

3. **Higher-Dimensional Carriers (S^n for $n \geq 3$):** The mathematical existence of higher-dimensional maximally symmetric topologies does not justify their ontological inclusion. The observable categories of relation are exhausted by directional shift (captured by the minimal 1-DOF S^1) and omnidirectional distribution (captured by the minimal 2-DOF S^2). Positing an S^3 or higher-dimensional carrier requires the existence of an independent, unobserved third category of macroscopic relation. By Principle 2.4 (Mathematical Transparency), introducing geometric degrees of freedom without uniquely identifiable physical relations constitutes over-parameterization. Furthermore, because higher-dimensional spheres contain S^1 and S^2 as lower-dimensional sub-structures, adopting them as foundational primitives when the minimal topologies suffice strictly violates Principle 2.3 (Ontological Minimalism). Therefore, all $n \geq 3$ carriers are excluded.

Ontological Minimalism and Mathematical Transparency enforce S^1 and S^2 as the minimal relational carriers without invoking metric geometry *a priori*. These two carriers are **sufficient** so far. Whether physical reality requires additional carriers beyond S^1 and S^2 is an open empirical question. □

- 1-DOF: minimum to encode ordered sequence (kinetic)
- 2-DOF: minimum to encode isotropic distribution (potential)
- 3-DOF+: reducible to combinations of above

WARNING: The Spatial Container Fallacy

S^1 and S^2 are not to be interpreted as physical geometries embedded in a spacetime container.
Do not visualise a sphere or a ring expanding through a 3D void.

- S^2 has no physical surface area, no volume, and does not "dilute" a flux across a spatial distance.
- They are purely abstract, algebraic phase spaces — relational carriers encoding the closure, conservation, and isotropy of the transformational resource.

Spatial distance (r) and time are purely emergent descriptive labels that observers attach to these underlying algebraic phase patterns. The phase projection dictates the space; the space does not dictate the phase.

Remark 3.2 (Constructive Derivation Protocol)**.** *This derivation proceeds by construction from traced origins. Each object entering the chain - the principles, the theorems, the carriers - has an explicit derivation path. Alternative structures (discrete groups, higher-dimensional carriers, non-orientable surfaces) are not excluded by enumeration; they are absent because no derivation chain within this framework produces them.*

3.3 The Amplitude-Phase Duality

S^1 and S^2 encode the *ratio* of internal to external relational energy (state transformation ledger 2.1.2). A point on S^1 and S^2 is not a position but a partition: how much of the system's transformation is externally expressed (amplitude β, κ) versus internally retained (phase β_Y, κ_X). The carrier geometry makes this partition exact and parameter-free.

Lemma 3.3 (Duality of Relation)**.** *By the Principle of Relational Origin (2.2), the physical manifestation of a state cannot be quantified by a single absolute magnitude. The elimination of background manifolds necessitates a complementary internal ratio of two measures:*

1. *the **amplitude** of transformation (external shift), and*

2. *the **phase** of transformation (internal order).*

Proof. By the Principle of Relational Origin (2.2), assigning a single unbounded scalar to describe a transformation introduces an unobservable absolute reference scale. A purely relational state must therefore be defined internally, as a ratio or partitioning of a closed resource.

The geometric carriers S^1 and S^2 mathematically enforce this requirement. Identifying a unique state on these closed carriers necessitates two orthogonal projections. A single projection cannot capture a ratio. One projection must quantify the extent of the external relation (Amplitude), which geometrically dictates an orthogonal complement quantifying the retained internal structure (Phase). Together, these dual, coupled axes provide the minimal, parameter-free ratio required to define a state transformation without invoking an external background. ☐

The orthogonal decomposition of the relational carriers S^1 and S^2 reveals a functional duality. Physical state is a combination of two projections:

Definition 3.4 (The Amplitude Projection (External Interaction))**.** *Denoted by β (kinematic) and κ (potential). This component measures the extent of external relation. It manifests as momentum or potential intensity. Physically, it represents the system's relational "shift" from the **Observer's Relational Origin** (the rest frame).*

$$\textit{Amplitude} \rightarrow \textit{External Power (Kinetic/Potential)}$$

Definition 3.5 (The Phase Projection (Internal Evolution))**.** *Denoted by β_Y and κ_X. This component measures the internal ordering. It governs the intrinsic scale of proper time and proper length. A Phase of 1 represents maximal internal flow (rest), while a Phase of 0 represents the collapse of internal causality (light-speed/event horizon).*

$$\textit{Phase} \rightarrow \textit{Internal Order (Time/Structure)}$$

Theorem 3.6 (Conservation of Relation)**.** *For both kinematic (S^1) and potential (S^2) modes, the sum of the squared Amplitude and squared Phase is invariant:*

$$\underbrace{\textit{Amplitude}^2}_{\textit{External Interaction}} + \underbrace{\textit{Phase}^2}_{\textit{Internal Existence}} = 1 \tag{1}$$

Proof. By Theorem 3.1, the balanced relational ledger is carried by the maximally symmetric geometries S^1 and S^2. Normalizing the total relational ledger to unity (1), any physical state corresponds to a point on these carriers. The algebraic identity of orthogonal projections on a unit circle and two-sphere dictates that the sum of their squares must equal the squared radius. Therefore, the parameters satisfy $\beta^2 + \beta_Y^2 = 1$ and $\kappa^2 + \kappa_X^2 = 1$, encoding the finite, closed relational ledger. ☐

3.3.1 Consequence: Relativistic Effects

Proposition 3.7 (Physical Interpretation: Relativistic Effects)**.** *The Conservation of Relation Theorem 3.6 implies that any redistribution between the orthogonal components manifests physically as the relativistic effects of time dilation and length contraction.*

Proof. By Theorem 3.6, the components satisfy $\beta^2 + \beta_Y^2 = 1$ and $\kappa^2 + \kappa_X^2 = 1$. An increase in Amplitude (β, κ) enforces a geometric decrease in Phase measure (β_Y, κ_X). This necessary reduction of β_Y and κ_X corresponds to the dilation of proper time and the contraction of proper length, while the growth of β and κ represents momentum or potential depth. Thus, the kinematic and potential trade-off is the direct, un-parameterized physical expression of the geometric closure of the S^1 and S^2 relational carriers. ☐

This identifies Relativistic and Gravitational Time Dilation not as a mysterious "slowing down" of clocks, but as geometric **phase rotation**. As a system invests more of its relational energy ledger into external Amplitude (β or κ), it necessarily withdraws from its internal Phase (β_Y or κ_X), changing the rate of its proper time evolution.

3.4 Phase Projections as Spectroscopic Shifts

Theorem 3.8 (Spectroscopic Phase Shift)**.** *The geometric projection κ^2 is determined by the measurable gravitational redshift z_κ via the identity:*

$$\kappa^2 = 1 - \frac{1}{(1+z_\kappa)^2}. \tag{2}$$

Proof. By the Conservation of Relation Theorem, the gravitational amplitude κ and internal phase κ_X satisfy:

$$\kappa^2 + \kappa_X^2 = 1 \quad \Longrightarrow \quad \kappa^2 = 1 - \kappa_X^2. \tag{3}$$

By Duality of Relation Lemma (3.3) the phase projection governs the intrinsic flow of proper time, the frequency of emitted light scales by the internal phase. The observable spectroscopic shift z_κ is therefore the operational ratio between the observer's phase and the source's phase:

$$1 + z_\kappa = \frac{\kappa_{X(obs)}}{\kappa_{X(source)}}. \tag{4}$$

Substituting the observer's maximal phase ($\kappa_{X(obs)} = 1$) — representing the invariance of the local measurement against the local frame — the phase of the source is isolated:

$$\kappa_{X(source)} = \frac{1}{1+z_\kappa}. \tag{5}$$

Substituting this internal phase back into the initial closure invariant yields the algebraic identity:

$$\boxed{\kappa^2 = 1 - \frac{1}{(1+z_\kappa)^2}} \ (z_\kappa = \text{gravitational redshift})$$

□

Theorem 3.9 (Kinematic Phase Shift (Transverse Doppler)). *The kinematic projection β^2 is determined by the transverse Doppler shift z_β via the symmetric identity:*

$$\beta^2 = 1 - \frac{1}{(1+z_\beta)^2}. \tag{6}$$

Proof. By the S^1 closure invariant (Conservation Theorem), the kinematic amplitude β and internal kinematic phase β_Y satisfy:

$$\beta^2 + \beta_Y^2 = 1 \quad \Longrightarrow \quad \beta^2 = 1 - \beta_Y^2. \tag{7}$$

The transverse Doppler shift z_β (without observer dependant line-of-sight kinematics) is the optical measure of phase discrepancy ratio:

$$1 + z_\beta = \frac{\beta_{Y(obs)}}{\beta_{Y(source)}}. \tag{8}$$

The observer's external kinematic projection is zero against local frame ($\beta_{obs} = 0$), yielding internal kinematic phase: $\beta_{Y(obs)} \equiv 1$. Substituting $\beta_{Y(obs)} = 1$, we isolate the internal phase of the moving source:

$$\beta_{Y(source)} = \frac{1}{1+z_\beta}. \tag{9}$$

Substituting this back into the S^1 closure invariant yields the algebraic identity:

$$\boxed{\beta^2 = 1 - \frac{1}{(1+z_\beta)^2}} \ (z_\beta = \text{transverse Doppler shift})$$

□

Theorem 3.10 (Operational Measurability). *The relational projections are encoded in the combined phase interactions of light (spectroscopy) and are operationally independent of spatial distance r, mass M, and the gravitational constant G.*

Proof. **Step 1: Total accumulated phase difference (τ).** Spectroscopy does not measure "gravitational potential" or "kinematic velocity" as independent isolated vectors; it measures the total accumulated phase discrepancy between the source and the observer. We define the **Relational Spacetime Factor** τ as the inverse of the system's combined measured redshift:

$$\tau \equiv \frac{1}{Z_{sys}} = \frac{1}{(1+z_\kappa)(1+z_\beta)}. \tag{10}$$

This single observable represents the product of the internal phase projections of the carriers S^2 and S^1:

$$\tau = \underbrace{\kappa_X}_{\text{Potential Phase}} \cdot \underbrace{\beta_Y}_{\text{Kinematic Phase}} = \sqrt{1-\kappa^2}\sqrt{1-\beta^2}. \tag{11}$$

Step 2: Relational Identity. Expanding the product of the squared phases yields the relationship between the optical signal τ and the geometric state:

$$\tau^2 = (1-\kappa^2)(1-\beta^2) = 1 - (\kappa^2+\beta^2) + \kappa^2\beta^2. \tag{12}$$

The optical signal contains the complete information regarding the system's structural relational state, acquired directly through observation without invoking absolute properties. □

3.5 Relational Origin of Spatial Distance

Spatial distance is derived as the inverse of the potential projection. The familiar inverse-square force law appears only in the metric scaffolding of legacy coordinate descriptions and is not part of the relational core.

Theorem 3.11 (Inverse-Distance Potential Amplitude)**.** *The inverse-distance potential proportionality $\kappa^2 \propto 1/r$ is a geometric necessity of the S^2 carrier's relational ledger, determined without spatial primitives.*

Proof. Distance cannot be an *a priori* foundational primitive. It violates the principles of Relational Origin (2.2) and Ontological Minimalism (2.3). Therefore it must be defined by the relational tension between energy states. By the Spectroscopic Phase Shift Theorem, the potential projection κ^2 is operationally extracted entirely from the dimensionless redshift z_κ, independent of spatial parameters.

S^2 is the omnidirectional relational carrier with two degrees of freedom. Let κ be the active potential projection per independent degree of freedom. Since the omnidirectional ledger distributes identically over both DOF (3.14), the dimensionless radial measure $\tilde{r}$ is the inverse of the external amplitude projection counted once for each DOF:

$$\tilde{r} = \frac{1}{\kappa} \cdot \frac{1}{\kappa} = \frac{1}{\kappa^2}. \tag{13}$$

Saturation point $\kappa^2 = 1$ defines causal horizon $r = R_s$ 3.12 and spatial distance emerges as

$$r = \frac{R_s}{\kappa^2}, \qquad \Longleftrightarrow \qquad \kappa^2 = \frac{R_s}{r}. \tag{14}$$

Thus $\kappa^2 \propto 1/r$. No background manifold, coordinate derivative, or force law is introduced. This is a direct realization of the Unifying Principle (2.13). The $1/r$ proportionality is an analytic identity defining spatial distance, not an empirical input. □

Remark 3.12. *Here and in all RG papers the causal horizon $\kappa = 1$ is symbolised with the usual letters for the Schwarzschild radius R_s for translational ease. Despite their ontological difference, these two horizons are mathematically identical. In order to keep the number of newly introduced symbols at a minimum, the historical symbols remain. The complete independence of causal horizon $\kappa = 1$ from its historical origin $R_s = \frac{2GM}{c^2}$ is proven in R.O.M. sections sec:unitless core, sec:relational_parameterization, sec:absolute_scale, sec:Kerr and others.*

Summary

The inverse-square proportionality does not exist because physical space dictates how forces dilute. It exists because "spatial distance" is the descriptive vocabulary assigned to the inverse amplitude of the potential relational carrier S^2.

The geometry of spacetime is dictated by the Relations of energy states.

The Projections are Cross-Cultural Invariants

- S^1 is the minimal geometric structure that can encode a directional relation without absolute space. Its parameter is an angle θ_1 whose cosine ($\beta = \cos(\theta_1)$) we identify with the fraction v/c and whose sine ($\beta_Y = \sin(\theta_1)$) governs the internal ordering (proper time fraction).
- S^2 is the minimal geometric structure that can encode omnidirectional relation without absolute space. Its parameter is an angle θ_2 whose sine ($\kappa = \sin(\theta_2)$) we identify with the escape velocity fraction v_e/c and whose cosine ($\kappa_X = \cos(\theta_2)$) governs the internal ordering (proper time fraction).

The unit radius of kinetic and potential carriers represents the Constant Universal Rate of Change, the Global Tempo of Transformations. The angles θ_1 and θ_2 control the distribution of this constant rate between the phase and amplitude components. *Any two observers regardless of their cultural differences (including hypothetical extraterrestrial intelligence) assign β and κ ratios the same value for the same state, whatever their unit system — or even their descriptive vocabulary. These mappings are NOT definitions:*

$$\beta = \frac{v}{c}, \qquad \kappa = \frac{v_e}{c} = \sqrt{\frac{R_s}{r}} = \sqrt{\frac{\rho}{\rho_{max}}}$$

They are *translations* into descriptive local historical vocabulary. Relational projections are **Metrologically Independent.**

3.6 Closure: Relational Origin of the Virial Theorem

Having established that directional (kinematic) and omnidirectional (potential) relations are carried by the S^1 and S^2 respectively, we now derive the relationship that unifies them.

Derivation of the Energetic Exchange Rate

Remark 3.13 (From slice to whole S^2)**.** *Although we parametrise a single meridional great circle (κ_X, κ) for algebraic convenience, the amplitude κ^2 denotes the* total active omnidirectional ledger *over the S^2 carrier. The exchange-rate factor 2 reflects that S^2 has two independent relational degrees of freedom even when calculations are carried out on a representative great-circle section.*

Uniqueness of the Exchange Rate (No Hidden Weighting)

Lemma 3.14 (DOF-Indifference)**.** *By maximal symmetry (no privileged directions, Theorem 2.10) and ontological minimalism (no hidden structure, Principle 2.3), each independent degree of freedom (DOF) has equal relational weight. WILL has no preferred directions and no preferred degrees of freedom.*

Proof. Any unequal weights to DOF's would create a preferred direction on maximally symmetric carriers violating the **Relational Isotropy Theorem** (2.10). It would require an extra rule that distinguishes the carriers on grounds other than their internal dimensions. Such a rule would constitute hidden structure, violating all four methodological principles (2.2, 2.3, 2.4, 2.1). Therefore, each independent DOF must receive an identical relational weight l. □

Theorem 3.15 (Closure)**.** *In a relationally closed system, the kinematic carrier S^1 (1 DOF) and the potential carrier S^2 (2 DOF) exchange the active relational ledger in proportion to their degrees of freedom. Consequently, any relationally closed system must globally satisfy*

$$\boxed{\kappa^2 = 2\beta^2}$$

Proof. By the Conservation Theorem (3.6) external amplitudes (β, κ) complete their ledger only in their quadratic forms ($\underbrace{\text{Amplitude}^2}_{\text{External Interaction}} + \underbrace{\text{Phase}^2}_{\text{Internal Existence}} = 1$). Let l be the active relational ledger allocated to one independent quadratic degree of freedom. By Lemma 3.14, every active quadratic DOF receives exactly this amount l.

- **Square amplitude on S^1 (1 DOF):** $\beta^2 = 1 \cdot l$
- **Square amplitude on S^2 (2 DOF):** $\kappa^2 = 2 \cdot l$
- **Substituting l immediately yields closure:** $\kappa^2 = 2\beta^2$

Any non-quadratic exchange law, same as any exchange rate other than 2 would require introduction of unjustified ontology either unequal weighting of degrees of freedom violating DOF-Indifference Lemma (3.14) or an additional free parameters violating all four principles (2.2, 2.3, 2.4, 2.1). Therefore:

$$\underbrace{1\text{Amplitude}^2}_{\text{on } S^2} = \underbrace{2\text{Amplitude}^2}_{\text{on } S^1}$$

□

Remark 3.16 (Carrier capacities fix the rate)**.** *The exchange rate 2 exactly mirrors the ratio of the carriers' maximum absolute capacities. The one-DOF carrier S^1 saturates at $\beta^2_{\max} = 1$ (light, $\beta_Y = 0$). The two-DOF carrier S^2 saturates at $\kappa^2_{\max} = 2$ (the combined extremal-Kerr state $r = \frac{1}{2}R_s$, $\beta = 1$ as proved in (R.O.M. sec:Kerr)). Equal quadratic weight per DOF (Lemma 3.14) guarantees that the active ledger ratio ($\kappa^2/\beta^2 = 2$) is identical to the maximal ratio ($\kappa^2_{\max}/\beta^2_{\max} = 2$).*

Definition 3.17 (Closure Factor)**.**

$$\delta \equiv \frac{\kappa_o^2}{2\beta_o^2}$$

Here and throughout all RG papers κ, β are global and κ_o, β_o are orbital-phase-dependent local amplitudes: $\kappa^2 = \langle\kappa_o^2\rangle$ and $\beta^2 = \langle\beta_o^2\rangle$ For circular orbits, $\delta \equiv 1$. For elliptic orbits locally, $\delta_{cycle} \neq 1$ as proven in R.O.M. eccentricity section . For bound elliptical orbits, $\langle\delta\rangle_{cycle} = 1$ to remain relationally closed *as proven in R.O.M. vis-viva section.*

Corollary 3.18 (Relational Closure Condition)**.** *Closed systems (momentary or periodic) satisfy $\kappa^2 = 2\beta^2$. Open systems display $\langle\delta\rangle_{cycle} \neq 1$, the magnitude of which quantifies the energy flow through unaccounted channels. When all channels are included, closure is restored.*

Illustrative Examples.

- **Circular Orbit (Closed).** A body at any orbital phase satisfies $\kappa_o^2 = 2\beta_o^2$. The entire active ledger is partitioned identically between kinetic and gravitational projections; no internal geometric oscillation and no external channel exists.
- **Elliptical Orbit (Closed).** A body satisfies $\langle\kappa_o^2\rangle = 2\langle\beta_o^2\rangle$ as an average per orbital cycle due to the internal geometric oscillation ("breathing") of eccentric systems. This internal oscillation is restricted by the Energy-Symmetry Law (3.10.2) so that the difference $\beta^2 = \kappa_o^2 - \beta_o^2 =$ constant at any orbital phase (R.O.M. sec:vis-viva). No external channel exists.

- **Radiating Binary (Open).** An elliptical compact binary violates $\langle\kappa_o^2\rangle = 2\langle\beta_o^2\rangle$ when only orbital degrees of freedom are counted. The closure defect δ quantifies the relational ledger lost to gravitational radiation. Including all channels restores closure.

Remark 3.19 (Geometric Origin of Physical Law)**.** *The relation between kinetic and potential energy projections is a geometric necessity of closure. Aspects of WILL maintain coherence through the invariant DOF ratio. This is the relational origin of the* ***virial theorem****:*

$$\boxed{\textit{Geometric Distribution } (\kappa^2) \;\equiv\; 2 \times \textit{Kinetic Distribution } (\beta^2).}$$

Structural Independence of the Force Law and Virial Coefficient

In classical mechanics, the virial coefficient is dynamically coupled to the spatial force law. For a power-law potential $V \propto r^n$, the general virial theorem dictates $2\langle T\rangle = (n+1)\langle V\rangle$. For Newtonian gravity ($n = -1$), this yields the coefficient 2. In this legacy view, the force law dictates the virial coefficient; they are two sides of the same dynamical coin.

WILL Relational Geometry decouples them. The $1/r$ proportionality and the virial coefficient 2 are proven to emerge from **entirely independent geometric channels**:

- **Channel 1 (Potential–Distance):** The $\kappa^2 \propto 1/r$ relation is derived strictly from the operational definition of spatial distance and the S^2 carrier's saturation geometry. The S^1 carrier and DOF ratios are entirely absent from this derivation.

- **Channel 2 (Potential–Kinetic Exchange):** The $\kappa^2 = 2\beta^2$ relation is derived strictly from the topological DOF ratio (1 for S^1, 2 for S^2) enforced by DOF-Indifference. Spatial distance r is entirely absent from this derivation.

Because both channels derive from the same underlying carrier topology, their mutual consistency is guaranteed *geometrically*, not dynamically. WILL precludes the possibility of a $1/r$ gravitational potential exhibiting a virial coefficient other than 2. This structural identity replaces the need for postulated equations of motion, Lagrangians, or action principles as foundational primitives.

Derivation Chain Map

Methodological constraints:
$\downarrow$(*Epistemic Hygiene+Relational Origin+Ontological Minimalism+Math Transparency*)$\downarrow$
Relational Closure + Causal Continuity + Isotropy Theorems
$\downarrow$ *(apply to existing physics)* $\downarrow$
Standard Primitives: $\underbrace{\text{background manifold + metric}}_{\text{spacetime structure (not derived)}} \quad + \quad \underbrace{\text{G, c, M, fields}}_{\text{energy dynamics (postulated)}}$
$\downarrow$ *(Ontological reduction by Relational Origin Principle — no background allowed)* $\downarrow$
One Primitive: ***WILL*** $\equiv$ ***SPACE-TIME-ENERGY*** $\Longrightarrow$ $\boxed{\text{SPACETIME} \equiv \text{ENERGY}}$
$\downarrow$ *(by Relational Closure + Causal Continuity + Isotropy: derive relational energy carriers)* $\downarrow$
S^1 **(1-DOF kinematic)** + S^2 **(2-DOF potential)**
$\downarrow$ *(by Duality of Relation Lemma)* $\downarrow$
Relational Conservation Theorem: $\underbrace{\text{Amplitude}^2}_{\text{External Transformation}} + \underbrace{\text{Phase}^2}_{\text{Internal Change}} = 1$
$\underbrace{1/\beta_Y = 1 + z_\beta}_{\text{transverse Doppler shift}}$ $\underbrace{\beta^2 + \beta_Y^2 = 1}_{\text{Kinematic } S^1}$ **Relational Projections** $\underbrace{\kappa^2 + \kappa_X^2 = 1}_{\text{Potential } S^2}$ $\underbrace{1/\kappa_X = 1 + z_\kappa}_{\text{gravitational redshift}}$
$\downarrow$ *(by DOF-Indifference Lemma: equal quadratic weight to each independent DOF)* $\downarrow$
Closure Theorem: $\underbrace{1\,\text{Amplitude}^2}_{\text{on } S^2} = \underbrace{2\,\text{Amplitude}^2}_{\text{on } S^1} \Longrightarrow \boxed{\kappa^2 = 2\beta^2}$

3.7 Kinetic Energy Projection β on S^1

- **The Amplitude Component** ($\beta = \cos(\theta_1) = v/c$)**:** External shift or the extent of kinematic relation observed from an arbitrary relational origin (the observer's rest frame). It quantifies how much of the universal "transformation rate" is perceived as motion through space, relative to the observer. In legacy units, this is identified as the dimensionless ratio of velocity to the speed of light.

- **The Phase Component ($\beta_Y = \sqrt{1-\beta^2}$):** Internal ordering or the intrinsic scale of proper time and proper length within the moving system. A phase of 1 signifies maximal internal flow (rest), while a phase of 0 indicates the collapse of internal causality (light-speed).

These two components are bound by the conservation theorem (3.6) in closed system, which acts as a strict "ledger of transformation":

$$\beta^2 + \beta_Y^2 = 1$$

The manifestation of any system is distributed between its internal (Phase) and relational (Amplitude) aspects.

Since S^1 encodes one-dimensional shift, the total energy E_β of the system must project consistently onto both external and internal axes:

$$E_\beta^2 = E_X^2 + E_Y^2$$

where

$$E_X = E_\beta \beta, \qquad E_Y = E_\beta \beta_Y.$$

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Kinetic projection β on the S^1 carrier

Geometric Nature of Rest Energy State

Theorem 3.20 (Invariant Projection of Rest Energy)**.** *For any state (β, β_Y) on the relational energy circle S^1, the total energy E_β must scale such that its internal projection remains constant and equal to the rest energy state E_0.*

$$E_\beta \beta_Y = E_0.$$

Proof. By the Relational Origin Principle (2.2), the internal structure of system A is determined by its rest frame ($\beta = 0, \beta_Y = 1$) as the rest energy E_0. When an external frame B observes system A in relative motion ($\beta > 0$), the geometric closure of the S^1 carrier necessitates a phase rotation ($\beta_Y < 1$). If the total observed energy E_β remained static ($E_\beta = E_0$), the projected internal energy $E_Y = E_\beta \beta_Y$ would decrease, implying that relative observation physically diminishes the internal energy of the observed system. This violates the structural integrity of the relation. To maintain the invariant internal reality of system A across all reference frames, the internal projection measured by B must perfectly equate to the rest state:

$$E_Y = E_\beta \beta_Y = E_0$$

Geometric consistency therefore dictates that the total energy E_β scales to compensate for the phase parameter:

$$E_\beta = \frac{E_0}{\beta_Y}$$

□

Summary:

The historical Lorentz factor γ is the reciprocal of β_Y. $\gamma = 1/\beta_Y$

The Geometric and Historical Nature of Mass

Corollary 3.21 (Rest Energy and Mass Equivalence)**.** *This geometry does not require mass as an independent concept. The invariant rest energy E_0 is numerically identical to mass m in natural units ($c = 1$), and is simply expressed in anthropocentric units (kilograms) via the conventional identity:*

$$E_0 = mc^2. \tag{15}$$

Proof. From the invariant projection $E_\beta \beta_Y = E_0$ and closure of S^1, no additional scaling parameter is required. Hence the conventional bookkeeping identities $E_0 = mc^2$ or $m = E_0/c^2$ reduce to tautologies. Mass is therefore not independent, but the rest-energy invariant itself. □

Summary:

Mass is the invariant projection of total rest energy expressed in anthropocentric units of kilograms.

Energy–Momentum Relation

Proposition 3.22 (Horizontal Projection as Momentum). *On the relational energy circle S^1, the unique projection of external amplitude from rest is the horizontal projection $E_X = E_\beta \beta$; hence*

$$p_\beta \equiv E_X \equiv E_\beta \beta \quad (c = 1).$$

Proof. A shift measure must (i) vanish at rest, (ii) grow monotonically with $|\beta|$, and (iii) flip sign under $\beta \mapsto -\beta$. As proven by Invariant Projection of Rest Energy Theorem 3.20 the rest state is $(\beta, \beta_Y) = (0, 1)$ and described by invariant vertical axis $E_Y = E_\beta \beta_Y = E_0$. The only relational candidate remaining and satisfying (i)-(iii) is the horizontal projection $E_X = E_\beta \beta$. Thus the identification is necessary and unique. □

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Energy-momentum Triangle

Corollary 3.23 (Energy–Momentum Relation). *With p identified by Proposition 3.22 and $m = E_0$, the closure identity yields*

$$E_\beta^2 = p_\beta^2 + m^2 \quad (c = 1).$$

Equivalently, upon restoring c,

$$E_\beta^2 = (p_\beta c)^2 + (mc^2)^2.$$

Proof. By Conservation Theorem (3.6), $(E_\beta \beta)^2 + (E_\beta \beta_Y)^2 = E_\beta^2$. Substituting $p_\beta = E_\beta \beta$ and $m = E_0$ proves the claim. Restoring c is dimensional bookkeeping: $p \mapsto pc$ and $m \mapsto mc^2$, while E_β remains E_β, yielding the standard form. Any surplus complexity is excluded by epistemic and ontological principles 2.1, 2.3. □

Remark 3.24 (Geometric Forms). *The same identity may be expressed explicitly in terms of circle coordinates:*

$$E_\beta^2 = (\tfrac{\beta}{\beta_Y} E_0)^2 + E_0^2 = \big(\cot(\theta_1)\, E_0\big)^2 + E_0^2.$$

These are equivalent renderings of the same geometric necessity.

Remark 3.25 (Units sanity check - bookkeeping). *In natural units ($c = 1$), the identification is $p_\beta \equiv E_\beta \beta$. Restoring c via dimensional bookkeeping ($p \mapsto pc$) yields:*

$$p_\beta c = E_\beta \beta = \frac{E_\beta}{c} v \implies p_\beta = \frac{E_\beta}{c^2} v. \tag{16}$$

With $E_\beta = \gamma mc^2$, this reduces to $p_\beta = \gamma mv$, the standard relativistic momentum. No new parameters are introduced.

$\beta = v/c \quad \theta_1 = \arccos(\beta)$	
Algebraic Form	**Trigonometric Form**
$\beta = v/c = \sqrt{1 - \beta_Y{}^2}$	$\beta = \cos(\theta_1)$
$\beta_Y = \sqrt{1 - (v/c)^2} = \sqrt{1 - \beta^2}$	$\beta_Y = \sin(\theta_1)$

Table 1: Geometric representation of relativistic effects.

Summary

The energy–momentum relation $E_\beta^2 = (p_\beta c)^2 + (mc^2)^2$ is geometric identity of S^1 relational energy carrier.

3.7.1 Minkowski Interval as Inflation of the S^1 Closure

In legacy Special Relativity the Minkowski interval is posited as the physical metric of a 4D background manifold. We reverse the order of explanation. The primitive object is the dimensionless kinematic closure on the relational carrier S^1 (Theorem 3.6):

$$\beta^2 + \beta_Y^2 = 1. \tag{17}$$

This is a single, parameter-free relation between the external (amplitude) and internal (phase) projections of the conserved transformation ledger. It invokes no coordinates, no container, and no external time. The Minkowski interval is *not* more fundamental than (17); it is what (17) becomes once a sequence of substantival posits is **added**, none of which is required by the relations themselves.

The inflationary posits. Beginning from the closure (17), legacy formalism introduces, in order:

(i) **A container.** The relation is declared to unfold *within* a pre-existing spatial manifold.

(ii) **Spatial coordinates.** Orthogonal labels x, y, z are imposed on the container to localize the amplitude projection.

(iii) **Externally flowing time.** An independent coordinate time t is posited and granted an autonomous flow, demoting the internal phase β_Y to a mere ratio of proper to coordinate time, $\beta_Y \equiv d\tau/dt$.

(iv) **A coordinate reference scale.** An **absolute temporal scale** c^2dt^2 as an external reference against which the dimensionless closure is measured. This is precisely the kind of unjustified background absolute that Relational Origin Principle 2.2 forbids.

The inflation. Under these posits the projections take their coordinate forms. The amplitude becomes the coordinate speed fraction distributed over the three container axes,

$$\beta^2 = \frac{v^2}{c^2} = \frac{dx^2 + dy^2 + dz^2}{c^2\, dt^2}, \tag{18}$$

and the phase becomes the time ratio $\beta_Y = d\tau/dt$. Substituting both into the closure (17),

$$\frac{dx^2 + dy^2 + dz^2}{c^2\, dt^2} + \left(\frac{d\tau}{dt}\right)^2 = 1, \tag{19}$$

multiplying both sides by c^2dt^2:

$$\frac{dx^2 + dy^2 + dz^2}{c^2dt^2} + \left(\frac{d\tau}{dt}\right)^2 = 1 \quad \xrightarrow{\times c^2dt^2} \quad dx^2 + dy^2 + dz^2 + c^2d\tau^2 = c^2dt^2$$

and clearing the posited denominator c^2dt^2 returns

Minkowski interval:

$$\underbrace{\beta^2 + \beta_Y^2 = 1}_{\text{irreducible primitive}} \quad + \quad \underbrace{x, y, z, t...}_{\text{coordinate inflation}} \quad = \underbrace{c^2d\tau^2 = c^2dt^2 - dx^2 - dy^2 - dz^2}_{\text{mathematical scaffolding}}$$

This is the Minkowski interval exactly. Shown as an inflation of simple relational conservation.

Epistemological Consequence: the Interval is the Inflated Form

The Minkowski interval is recovered without remainder — but only after four substantival structures (a container, three spatial coordinates, an autonomously flowing time, and a coordinate reference scale) have been appended to $\beta^2 + \beta_Y^2 = 1$. Not one of these is demanded by the relations; each is surplus ontology. The Lorentz factor $\gamma = 1/\beta_Y = 1/\sin\theta_1$ and kinematic time dilation ($\beta_Y < 1$) are accordingly not deformations of a spatial fabric but a phase rotation on the unit circle S^1. The interval carries more structure than the relation it encodes; the relation, not the interval, is the irreducible fundamental primitive.

3.8 Potential Energy Projection κ on S^2

IMPORTANT:

Throughout this work, S^1 and S^2 are not to be interpreted as spacetime geometries but purely as relational carriers encoding energy conservation. Any reading otherwise is a misinterpretation.

Analogous to S^1, the relational geometry of the sphere S^2 provides orthogonal projections for two aspects of omnidirectional transformation. We define them as follows:

- **The Amplitude Component ($\kappa = \sin(\theta_2) = \frac{v_e}{c} = \sqrt{\frac{R_s}{r}}$ 3.12):** The *relational potential measure* between the object and the observer. It corresponds to the *extent* of transformation as potential depth. Same way as we did with β on S^1, we will map κ on S^2 as fraction of speed of light. A value of $\kappa = 1$ denotes *saturation*: the entire relational resource of the system has been allocated into the potential channel. This condition mathematically defines the relational horizon where the escape velocity is equal to speed of light ($v_e = c$) or equivalently as radial distance equal to Schwarzschild radius ($r = R_s$ event horizon).
- **The Phase Component ($\kappa_X = \sqrt{1 - \kappa^2}$):** This projection governs the intrinsic scale of proper length and proper time units within the potential field, corresponding to the internal *sequence* of transformation.

These two components are bound by the conservation theorem (3.6) in closed system, which acts as a finite "budget of transformation":

$$\kappa_X^2 + \kappa^2 = 1$$

The manifestation of any system is distributed between its internal (Phase) and relational (Amplitude) aspects. This single geometric constraint gives rise to the core gravitational phenomena of modern physics.

Remark 3.26. *The additive closure $\kappa_X^2 + \kappa^2 = 1$ holds for static systems; rotation promotes the system to the composite product phase $\tau^2 = \beta_Y^2 \kappa_X^2 = (1-\beta^2)(1-\kappa^2)$ lifting the $\kappa \leq 1$ bound as we show in Kerr section.*

Desmos project

Potential projection as κ on S^2 carrier

Potential Meridional Section of S^2

By Isotropy Theorem 2.10, the omnidirectional energy carrier is S^2, but any radially symmetric relation reduces to a great-circle meridional section. We therefore work on representative unit great circle S^2 without loss of generality with the parameterization $(\kappa_X, \kappa) = (\cos\theta_2, \sin\theta_2)$.

Consequence: Gravitational Effects

The redistribution of the relational ledger between the Phase and Amplitude components directly produces the effects of General Relativity. An increase in the relational amplitude measure (κ, gravitational potential) strictly necessitates a geometric decrease in the measure of the internal structure (κ_X). This geometric trade-off is observed physically as gravitational length contraction and time dilation. Thus, the curved geometry of spacetime is merely the shadow cast by the geometry of conserved relations.

Gravitational Tangent Formulation

Just as the relativistic energy–momentum relation can be expressed in terms of the kinematic projection $\beta = v/c$, we may construct its gravitational analogue using the potential projection $\kappa = v_e/c$, where v_e is the Newtonian escape velocity at radius r.

In the kinematic case, with $\beta = \cos\theta_1$, the energy relation can be written as

$$E_\beta^2 = \left(\cot\theta_1\, E_0\right)^2 + E_0^2, \tag{20}$$

so that the relativistic momentum is expressed as

$$p_\beta = (E_0/c)\cot\theta_1. \tag{21}$$

In full geometric symmetry, the gravitational case follows from $\kappa = \sin\theta_2$. We define the scalable gravitational total energy as

$$E_\kappa = \frac{E_0}{\kappa_X}, \qquad \kappa_X = \sqrt{1-\kappa^2}, \tag{22}$$

and introduce the gravitational analogue of momentum (potential momentum):

$$p_\kappa \ = \ (E_0/c)\tan\theta_2. \tag{23}$$

This yields the exact gravitational energy–momentum relation:

$$E_\kappa^2 = (p_\kappa c)^2 + (mc^2)^2. \tag{24}$$

Remark 3.27 (Ontological Status of Gravitational Momentum (p_κ))**.** *The gravitational momentum p_κ does not constitute a new empirical observable distinct from standard kinematic measurements; rather, it is the* ***ontological reinterpretation*** *of gravitational energy-momentum within the relational ontology.*

- ***Empirical Grounding:*** *p_κ is defined by the operationally measurable gravitational redshift z_κ, exactly as kinematic momentum p_β is defined by the transverse Doppler shift z_β (3.4).*
- ***Geometric Equivalence Principle:*** *For a system in radial free-fall, the kinematic and potential amplitudes equalize ($\beta = \kappa$), yielding $p_\beta = p_\kappa$. This numerical identity is not a postulated equivalence, but a geometric necessity derived from the parallel closures of S^1 and S^2 and confirmed by Unified Relational Scaling Lemma (4.1).*
- ***Resolution of the Pseudotensor Problem:*** *Standard GR fails to localize gravitational energy because it treats energy as a property of a spatial region (the metric field). Because p_κ is relational measure between physical observers (quantified by spectroscopic phase shift), it requires no spatial container. It replaces the unlocalizable gravitational pseudotensor of GR with a clean, operationally defined relational ledger.*

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Summary:

$$\beta = \cos\theta_1, \qquad \kappa = \sin\theta_2,$$
$$\beta \longleftrightarrow \kappa, \qquad \cot\theta_1 \longleftrightarrow \tan\theta_2.$$

Kinematic momentum p and gravitational momentum p_κ are thus dual projections of the closed relational geometry, expressed through complementary trigonometric forms.

3.8.1 Schwarzschild Interval as Inflation of the S^2 Closure

The gravitational construction is parallel to the kinetic. The primitive object is the dimensionless potential closure on the relational carrier S^2 (Theorem 3.6):

$$\kappa^2 + \kappa_X^2 = 1, \tag{25}$$

relating the external potential amplitude κ to the internal phase κ_X. As before, (25) contains no radial coordinate, no central body, and no external time. The Schwarzschild component g_{tt} is recovered only by appending substantival posits.

The inflationary posits.

(i) **A container.** A spatial manifold is posited around a central body.

(ii) **A static observer.** The observer is held fixed in this container ($dr = d\theta = d\phi = 0$), so that only the temporal relation survives.

(iii) **Externally flowing time.** An independent coordinate time t is posited with autonomous flow, demoting the internal phase to the ratio $\kappa_X \equiv d\tau/dt$.

(iv) **A radial coordinate and legacy constants.** The dimensionless amplitude is localized as a coordinate function, $\kappa^2 \equiv R_s/r$ 3.12, thereby introducing a radial coordinate r together with the legacy constants G and M through $R_s = 2GM/c^2$ — none of which is intrinsic to κ.

The inflation. Substituting $\kappa_X = d\tau/dt$ and $\kappa^2 = R_s/r$ 3.12 into the closure (25),

$$\left(\frac{d\tau}{dt}\right)^2 + \frac{R_s}{r} = 1 \quad \Longrightarrow \quad \left(\frac{d\tau}{dt}\right)^2 = 1 - \frac{R_s}{r}, \tag{26}$$

and introducing the posited reference scale c^2dt^2 gives

Schwarzschild interval:

$$\underbrace{\kappa^2 + \kappa_X^2 = 1}_{\text{irreducible primitive}} \quad + \quad \underbrace{x, y, z, t...}_{\text{coordinate inflation}} \quad = \underbrace{c^2d\tau^2 = c^2\left(1 - \frac{R_s}{r}\right)dt^2}_{\text{mathematical scaffolding}},$$

which is the Schwarzschild interval for a static observer ($ds^2 = c^2d\tau^2$).

Remark 3.28 (Full derivation)**.** *This covers the static interval. The full derivation of the dynamic Schwarzschild and Kerr metrics is presented in the supplement: R.O.M. (sec:Schwarzschild), R.O.M. (sec:Kerr).*

Two Inflations

The Minkowski and Schwarzschild intervals are produced by the identical procedure: append a container, coordinates, an autonomously flowing time, and a coordinate reference scale to a single dimensionless closure — $\beta^2 + \beta_Y^2 = 1$ on the kinematic carrier, $\kappa^2 + \kappa_X^2 = 1$ on the potential carrier. Gravitational time dilation ($\kappa_X < 1$) is, like its kinematic counterpart, a phase rotation, not the curvature of a pre-existing fabric. What General Relativity treats as the fundamental metric of spacetime is the coordinate-laden image of a parameter-free relation. The relation is primary; the metric is its inflation.

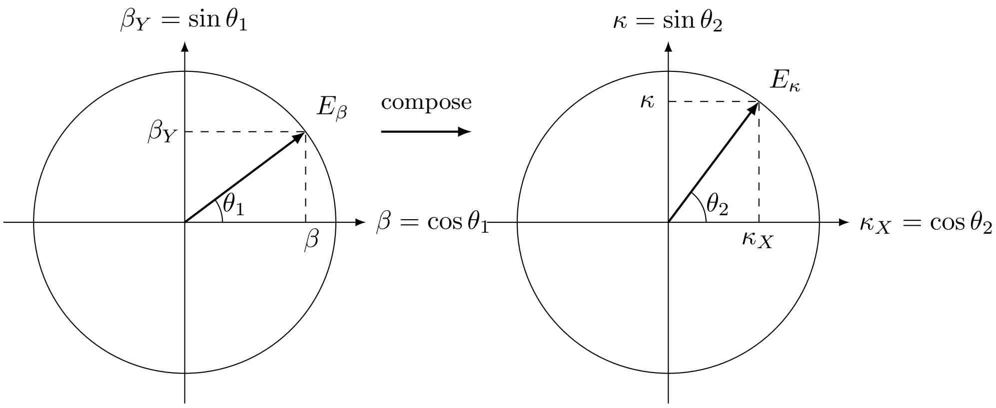
Minkowski interval:

$$\underbrace{\beta^2 + \beta_Y^2 = 1}_{\text{irreducible primitive}} \quad + \quad \underbrace{x, y, z, t...}_{\text{coordinate inflation}} \quad = \underbrace{c^2d\tau^2 = c^2dt^2 - dx^2 - dy^2 - dz^2}_{\text{mathematical scaffolding}}$$

Schwarzschild interval:

$$\underbrace{\kappa^2 + \kappa_X^2 = 1}_{\text{irreducible primitive}} \quad + \quad \underbrace{x, y, z, t...}_{\text{coordinate inflation}} \quad = \underbrace{c^2d\tau^2 = c^2\left(1 - \frac{R_s}{r}\right)dt^2}_{\text{mathematical scaffolding}},$$

3.9 Clear Relational Symmetry Between Kinematic and Potential Projections



$E_\beta = E_0/\beta_Y$
$p = E_0\beta/\beta_Y = E_0 \cot\theta_1$

$E_\kappa = E_0/\kappa_X$
$p_\kappa = E_0\kappa/\kappa_X = E_0 \tan\theta_2$

On the unit kinematic circle (S^1) we parameterize

$$(\beta, \beta_Y) = (\cos\theta_1,\ \sin\theta_1),$$

so that the invariant internal phase projection reads

$$E_\beta\, \beta_Y = E_0 \Longrightarrow \boxed{E_\beta = \frac{E_0}{\beta_Y} = \frac{E_0}{\sin\theta_1}}, \qquad p_\beta = \frac{E_\beta}{c}\,\beta = \frac{E_0\,\beta}{\beta_Y} = E_0 \cot\theta_1,$$

and therefore $E_\beta^2 = (p_\beta c)^2 + E_0^2$.

On the unit potential 2-sphere (S^2) we parameterize

$$(\kappa_X, \kappa) = (\cos\theta_2,\ \sin\theta_2),$$

so that the invariant internal phase projection reads

$$E_\kappa\, \kappa_X = E_0 \Longrightarrow \boxed{E_\kappa = \frac{E_0}{\kappa_X} = \frac{E_0}{\cos\theta_2}}, \qquad p_\kappa = E_\kappa\, \kappa = \frac{E_0\,\kappa}{\kappa_X} = E_0 \tan\theta_2,$$

and therefore $E_\kappa^2 = (p_\kappa c)^2 + E_0^2$.

Now we can clearly see the underlying symmetry between relativistic and gravitational factors that can be expressed in unified algebraic and trigonometric forms, as shown in Table 2.

$\theta_1 = \arccos(\beta)$, $\theta_2 = \arcsin(\kappa)$, $\kappa^2 = 2\beta^2$, $a =$ semi-major axis	
Algebraic Form	**Trigonometric Form**
$\beta = v/c$	$\beta = \cos(\theta_1)$
$\kappa = \sqrt{R_s/a}$	$\kappa = \sin(\theta_2)$
$\beta_Y = \sqrt{1-\beta^2}$	$\beta_Y = \sin(\theta_1)$
$\kappa_X = \sqrt{1-\kappa^2}$	$\kappa_X = \cos(\theta_2)$
$E_\beta = E_0 \cdot 1/\beta_Y$	$E_\beta = E_0 \cdot \csc(\theta_1)$
$E_\kappa = E_0 \cdot 1/\kappa_X$	$E_\kappa = E_0 \cdot \sec(\theta_2)$
$E = E_0 \cdot \kappa_X/\beta_Y$	$E = E_0 \cdot \cos(\theta_2)\csc(\theta_1)$
$p_\beta c = E_0 \cdot \beta/\beta_Y$	$p_\beta c = E_0 \cdot \cot(\theta_1)$
$p_\kappa c = E_0 \cdot \kappa/\kappa_X$	$p_\kappa c = E_0 \cdot \tan(\theta_2)$
$\tau = \beta_Y \kappa_X$	$\tau = \sin(\theta_1)\cos(\theta_2)$
$\tau_Y = \sqrt{1-\tau^2}$	$\tau_Y = \sqrt{1-\tau^2}$

Table 2: Unified representation of relativistic and gravitational effects for relationally closed systems .

Summary

The familiar SR and GR factors emerge here as projections of the same conserved resource. Relativistic (β) and gravitational (κ) modes are not separate "effects" but dual aspects of one energy-transformation constraint revealing their unified origin.

3.10 Energy Symmetry and Causal Continuity

3.10.1 Three Relational Energy Quantities

The formalism employs three distinct energies. Confusing them is the most common source of ambiguity, so we define and relate them explicitly. None of them presupposes a background frame; each is defined purely from the carrier closures.

- **Kinematic carrier energy** E_β**:** From the invariant rest energy projection theorem (3.20), $E_\beta\,\beta_Y = E_0$, so $E_\beta = E_0/\beta_Y = E_0/\sin\theta_1$. This is the energy of a state whose only shift is kinematic ($\kappa = 0$). It reduces to the rest energy at $\beta = 0$ and diverges as $\beta \to 1$ (the speed-of-light horizon). It is the relational analogue of γmc^2.
- **Potential carrier energy** E_κ**:** From the analogous invariant projection (Sec. 3.8), $E_\kappa\,\kappa_X = E_0$, so $E_\kappa = E_0/\kappa_X = E_0/\cos\theta_2$. This is the energy of a state whose only shift is gravitational ($\beta = 0$). It reduces to the rest energy at $\kappa = 0$ and diverges as $\kappa \to 1$ (the causal horizon). Like E_β, it is a boosted carrier energy; it is not the energy of the combined state.
- **Relational total energy** E**:** When both carriers are active, the single conserved energy of the relation — the quantity the Energy-Symmetry Law balances — is the phase-ratio scaling $E = E_0\,\kappa_X/\beta_Y$.

The three are related by $E = E_0\,\kappa_X/\beta_Y = \kappa_X \cdot E_\beta = E_0 \cdot E_\beta/E_\kappa$, i.e. the relational energy is the kinematic carrier energy E_β scaled by the potential phase κ_X (equivalently, E_0 times the ratio of the two carrier energies). The relational total energy E is the mixed quantity conserved by Causal Continuity Theorem (2.6).

The limiting behavior is immediate: in the "flat" / no-gravity limit ($\kappa \to 0$), $E \to E_0/\beta_Y = E_\beta$ (pure kinematics); in the "static" / no relative motion limit ($\beta = 0$), $E = E_0\,\kappa_X = E_0\sqrt{1 - R_s/r}$, the familiar gravitational redshift factor. In this same limit the carrier energy is $E_\kappa = E_0/\kappa_X$, so $E \cdot E_\kappa = E_0^2$: the relational energy and the asymptotic carrier energy are reciprocal duals.

3.10.2 Causal Continuity and Energy Symmetry

Theorem 3.29 (Energy Symmetry)**.** *Any closed system conserves its relational energy ledger with an invariant zero sum of total transformations. Transitions between states of observers (per unit of rest energy) must always sum to zero. Nothing is created, nothing is destroyed:*

$$\Delta E_{A\to B} + \Delta E_{B\to A} = 0. \tag{27}$$

Proof. By Theorems **Relational Closure** (2.5) and **Causal Continuity** (2.6), the energy of a system is determined by the ratio of the potential phase to the kinematic phase:

$$E = E_0 \frac{\kappa_X}{\beta_Y} \tag{28}$$

Consider two observers, A and B. The specific energy difference perceived for a transition between their states is determined by the difference in their total phase scaling:

$$\Delta E_{A\to B} = E_0 \left(\frac{\kappa_{X,B}}{\beta_{Y,B}} - \frac{\kappa_{X,A}}{\beta_{Y,A}} \right) \tag{29}$$

Conversely, the specific energy difference perceived for the reverse transition is:

$$\Delta E_{B\to A} = E_0 \left(\frac{\kappa_{X,A}}{\beta_{Y,A}} - \frac{\kappa_{X,B}}{\beta_{Y,B}} \right) \tag{30}$$

Summing these transfers gives:

$$\Delta E_{A\to B} + \Delta E_{B\to A} = 0 \tag{31}$$

Thus, the Energy-Symmetry Law emerges directly from the phase ratio difference, ensuring causal continuity and unbroken ledger across all transitions. □

Theorem 3.30 (Universal Rate of Change and the Horizon of Causality)**.** *What we perceive as the speed of light or event horizon is the exhaustion of the phase, as the constant rate becomes fully distributed onto the amplitude axis. In legacy units this takes form of the universal speed limit ($v \le c$) and the Schwarzschild limit ($r \ge R_s$). In relational ontology this limits emerge from the requirement of energetic symmetry as natural boundaries of causal connection.*

Proof. The unit radius of the kinetic carrier S^1 and the potential carrier S^2 represents the Constant Universal Rate of Change — the Global Tempo of Transformations. The closure identities $\beta^2 + \beta_Y^2 = 1$ and $\kappa^2 + \kappa_X^2 = 1$ dictate the distribution of this constant rate between the amplitude (external interaction) and the phase (internal existence).

The total local energy ledger is scaled by the ratio of the potential phase to the kinematic phase: $E = E_0 \frac{\kappa_X}{\beta_Y}$. The Energy-Symmetry Law $\Delta E_{A\to B} + \Delta E_{B\to A} = 0$ requires that any transition between states possesses a finite, symmetric counter-transition to close the ledger.

$$\lim_{\beta_Y \to 0} E \to \infty \quad \text{and} \quad \lim_{\kappa_X \to 0} E \to 0 \tag{32}$$

Kinematic Phase Exhaustion (S^1) Speed of light:
$\beta \to 1 \ \to \ \beta_Y \to 0 \ \to \ E = E_0 \cdot \kappa_X/0 \to \infty \ \to$ **requires** ∞ **energy**
As a system distributes its Global Transformation Tempo increasingly into external kinematic interaction, the amplitude β approaches 1. To maintain the carrier's closure, the internal kinematic phase approaches exhaustion: $\beta_Y \to 0$.
The local energy scaling diverges: $E = E_0 \frac{\kappa_X}{0} \to \infty$.
To preserve the Energy-Symmetry Law, the balancing reverse transition must supply an infinite relational counter-transfer. Because finite physical systems possess finite relational ledger, an infinite counterpart is structurally unattainable. The ledger cannot be closed. Therefore, $\beta_Y = 0$ constitutes a strict causal horizon—the Universal Rate of Change limit ($v = c$)—where energetic symmetry prohibits further amplitude growth.

Potential Phase Exhaustion (S^2) Event Horizon:
$\kappa \to 1 \ \to \ \kappa_X \to 0 \ \to \ E = E_0 \cdot 0/\beta_Y \to 0 \ \to$ **requires** 0 **energy**
As a system distributes its Global Tempo increasingly into external potential depth, the amplitude κ approaches 1. The internal potential phase approaches exhaustion: $\kappa_X \to 0$.
The local energy scaling collapses: $E = E_0 \frac{0}{\beta_Y} \to 0$.
To preserve the Energy-Symmetry Law, any transition across this boundary requires a counterpart with zero energy to balance the ledger. Consequently, no finite, non-zero transformation can symmetrically cross this threshold. The point $\kappa_X = 0$ constitutes the relational causal horizon (the Schwarzschild radius), where the internal phase ledger is fully depleted into external potential amplitude, isolating the system from finite causal interactions.

Thus, the universal speed limit and the Event Horizon emerge identically as the necessary causal horizons of Relational Geometry, enforcing the invariant symmetry of the energy ledger against phase exhaustion. □

Summary

Energy Symmetry Law $\Delta E_{A\to B} + \Delta E_{B\to A} = 0$ is the universal relational energy bookkeeping. **The Speed of Light and the Schwarzschild radius** are legacy units expressions of the boundaries beyond which the energy symmetry law breaks down. **Causality** is an **inevitable** feature of **Relational Geometry**.

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Derivation Chain Map

Methodological constraints:
↓(*Epistemic Hygiene+Relational Origin+Ontological Minimalism+Math Transparency*)↓
Relational Closure + Causal Continuity + Isotropy Theorems
↓ *(apply to existing physics)* ↓
Standard Primitives: $\underbrace{\text{background manifold + metric}}_{\text{spacetime structure (not derived)}} \quad + \quad \underbrace{\text{G, c, M, fields}}_{\text{energy dynamics (postulated)}}$
↓ *(Ontological reduction by Relational Origin Principle — no background allowed)* ↓
One Primitive: ***WILL*** $\equiv$ ***SPACE-TIME-ENERGY*** $\Longrightarrow$ $\boxed{\text{SPACETIME} \equiv \text{ENERGY}}$
↓ *(by Relational Closure + Causal Continuity + Isotropy: derive relational energy carriers)* ↓
S^1 **(1-DOF kinematic) +** S^2 **(2-DOF potential)**
↓ *(by Duality of Relation Lemma)* ↓
Relational Conservation Theorem: $\underbrace{\text{Amplitude}^2}_{\text{External Transformation}} + \underbrace{\text{Phase}^2}_{\text{Internal Change}} = 1$
$\boxed{\underbrace{1/\beta_Y = 1 + z_\beta}_{\text{transverse Doppler shift}} \quad \underbrace{\beta^2 + \beta_Y^2 = 1}_{\text{Kinematic } S^1}}$ **Relational Projections** $\boxed{\underbrace{\kappa^2 + \kappa_X^2 = 1}_{\text{Potential } S^2} \quad \underbrace{1/\kappa_X = 1 + z_\kappa}_{\text{gravitational redshift}}}$
↓ *(by DOF-Indifference Lemma: equal quadratic weight to each independent DOF)* ↓
Closure Theorem: $\underbrace{1\,\text{Amplitude}^2}_{\text{on } S^2} = \underbrace{2\,\text{Amplitude}^2}_{\text{on } S^1} \Longrightarrow \boxed{\kappa^2 = 2\beta^2}$
↓ *(by Relational Conservation Theorem derive metric intervals as scaffolding)* ↓
Minkowski: $\underbrace{\beta^2 + \beta_Y^2 = 1}_{\text{irreducible primitive}} + \underbrace{x, y, z, t...}_{\text{coordinate inflation}} = \underbrace{c^2 d\tau^2 = c^2 dt^2 - dx^2 - dy^2 - dz^2}_{\text{mathematical scaffolding}}$
Schwarzschild: $\underbrace{\kappa^2 + \kappa_X^2 = 1}_{\text{irreducible primitive}} + \underbrace{x, y, z, t...}_{\text{coordinate inflation}} = \underbrace{c^2 d\tau^2 = c^2 (1 - (R_s/r))\, dt^2}_{\text{mathematical scaffolding}}$
↓ *(by Relational Closure and Causal Continuity Theorems: any change must be balanced)* ↓
Energy Symmetry Law: $\boxed{\Delta E_{A\to B} + \Delta E_{B\to A} = 0}$
↓ $\Delta E_{A\to B} = E_0(\kappa_{X,B}/\beta_{Y,B} - \kappa_{X,A}/\beta_{Y,A})$; $\Delta E_{B\to A} = E_0(\kappa_{X,A}/\beta_{Y,A} - \kappa_{X,B}/\beta_{Y,B})$ ↓
Speed of light: $\beta \to 1 \;\to\; \beta_Y \to 0 \;\to\; E = E_0 \cdot \kappa_X/0 \to \infty \;\to$ **requires** ∞ **energy**
Event Horizon: $\kappa \to 1 \;\to\; \kappa_X \to 0 \;\to\; E = E_0 \cdot 0/\beta_Y \to 0 \;\to$ **requires** 0 **energy**

4 Stage III: Legacy Translation & Empirical Alignment

4.1 Equivalence Principle as Derived Identity

Lemma 4.1 (Unified Relational Scaling)**.** *Within relational ontology, both kinematic (S^1) and potential (S^2) transformations act as projections of the same invariant rest energy E_0. The total local energy scale is determined by the ratio of the potential phase to the kinematic phase:*

$$E = E_0 \frac{\kappa_X}{\beta_Y} = E_0 \sqrt{\frac{1-\kappa^2}{1-\beta^2}}.$$

Proof. On the kinematic circle S^1, the internal phase corresponds to $\beta_Y = \sin\theta_1$. On the potential sphere S^2, the internal phase is $\kappa_X = \cos\theta_2$. The total local energy is defined by the ratio of these internal phases, scaling the invariant rest energy E_0. This scaling preserves the closure identity of the respective carriers and governs all external manifestations of the system's energy. □

Theorem 4.2 (Equivalence of Inertial and Gravitational Response)**.** *The total local energy scale is given by the phase ratio:*

$$E = E_0 \frac{\kappa_X}{\beta_Y} = E_0 \sqrt{\frac{1-\kappa^2}{1-\beta^2}}.$$

The corresponding inertial and gravitational projections share a single operational factor,

$$p_\beta = \frac{E}{c}\beta, \qquad p_\kappa = \frac{E}{c}\kappa,$$

both governed by the same effective mass scaling

$$m_{\text{eff}} = \frac{E}{c^2} = \frac{E_0\kappa_X}{\beta_Y c^2}.$$

Therefore, the underlying invariant mass for both projections is identical:

$$\boxed{m_g \equiv m_i \equiv m = E_0/c^2},$$

and the Einstein equivalence principle follows as a necessary structural identity of WILL Relational Geometry.

Proof.

- **Inertial mass** is operationally defined as resistance to kinematic acceleration — it appears in $p_\beta = m_{\text{eff}} \cdot \beta c$.
- **Gravitational mass** is operationally defined as coupling to gravitational potential — it appears in $p_\kappa = m_{\text{eff}} \cdot \kappa c$.
- **Both projections** share the identical effective prefactor $m_{\text{eff}} = \frac{E}{c^2}$ because both p_β and p_κ are projections of the same conserved internal rest state E_0 ($\beta_Y = \kappa_X = 1$). Relational phase factors β_Y and κ_X rescale only external manifestations (effective energy and momentum), while the base invariant E_0 remains unchanged.
- **The equivalence** is the geometric consequence of p_β and p_κ being dual projections of a single relational rest state, requiring no independent mass parameters:

$$\boxed{m_g \equiv m_i \equiv m = E_0/c^2}$$

□

Remark 4.3 (Composition-Independence)**.** *Decomposing the invariant rest energy into internal channels,*

$$E_0 = \sum_a E_0^{(a)},$$

each term couples identically through the same phase ratio scaling:

$$E = \sum_a \left(E_0^{(a)} \frac{\kappa_X}{\beta_Y} \right).$$

Since all channels scale by the same phase ratio $\frac{\kappa_X}{\beta_Y}$, ratios between channels cancel in all observables. Therefore, composition-independence of motion (Eötvös universality) follows identically, without requiring a postulate $m_g = m_i$.

4.2 Classical Mechanics as Collapsed Two-Point Relational Ontology

4.2.1 The Phase Scaling and the Classical $\frac{1}{2}$

The total energy in RG is given by the phase ratio:

$$\frac{E}{E_0} = \frac{\kappa_X}{\beta_Y} = \frac{\sqrt{1-\kappa^2}}{\sqrt{1-\beta^2}} = (1-\kappa^2)^{1/2}(1-\beta^2)^{-1/2} \tag{33}$$

In the weak-field, low-velocity regime ($\beta \ll 1$, $\kappa \ll 1$), the Taylor expansion of the phase scaling yields:

$$\frac{E}{E_0} \approx \left(1 - \frac{1}{2}\kappa^2\right)\left(1 + \frac{1}{2}\beta^2\right) \approx 1 - \frac{1}{2}\kappa^2 + \frac{1}{2}\beta^2 \tag{34}$$

The excess energy (the kinetic and potential ledgers) per unit rest energy is therefore:

$$\frac{E - E_0}{E_0} \approx \frac{1}{2}\beta^2 - \frac{1}{2}\kappa^2 \tag{35}$$

Relational Origin of the Classical $\frac{1}{2}$

The historical factor $\frac{1}{2}$ in classical kinetic and potential energy formulas is the first-order Taylor coefficient of the phase ratio κ_X/β_Y. Newtonian physics emerges as the linear approximation of the relational phase ratio. Classical mechanics captures the first-order, a testament to its empirical adequacy in the weak-field regime.

Physical Interpretation

This demonstrates that the Keplerian total energy is a direct linear projection of a deeper relational structure. The difference of squared amplitudes ($\beta^2 - \kappa^2$) emerges invariantly from the closed geometry of the carriers. The structure of the energy relation is a direct consequence of the relational closure of S^1 and S^2.

4.2.2 Lagrangian and Hamiltonian

In the standard formulation of mechanics, the Lagrangian $L = T - V$ and the Hamiltonian $H = T + V$ are treated as fundamental functions of a single configuration point $\mathbf{r}$. Relational Geometry reveals that both are linearized limits of a deeper two-point energy balance: the **Energy-Symmetry Law**

$$\Delta E_{A\to B} + \Delta E_{B\to A} = 0,$$

where the specific energy (per unit rest energy) perceived by an observer at state A for a transition to state B is determined by the difference in their total phase scaling:

$$\boxed{\Delta E_{A\to B} = \left(\frac{\kappa_X}{\beta_Y}\right)_B - \left(\frac{\kappa_X}{\beta_Y}\right)_A}. \tag{36}$$

Here $\beta = v/c$ is the kinematic projection on the carrier S^1 and $\kappa^2 = R_s/r$ is the potential projection on the carrier S^2.

The linearized relational limit

To connect with legacy mechanics, we examine the weak-field, low-velocity regime ($\beta \ll 1, \kappa \ll 1$). The Taylor expansion of the phase ratio yields $\frac{\kappa_X}{\beta_Y} \approx 1 - \frac{1}{2}\kappa^2 + \frac{1}{2}\beta^2$. Substituting this into the two-point law (Eq. 36) gives:

$$\Delta E_{A\to B} \approx \left(1 - \tfrac{1}{2}\kappa_B^2 + \tfrac{1}{2}\beta_B^2\right) - \left(1 - \tfrac{1}{2}\kappa_A^2 + \tfrac{1}{2}\beta_A^2\right).$$

The constant 1 cancels out, leaving the linearized specific energy transfer:

$$\Delta E_{A\to B} \approx \tfrac{1}{2}\left(\kappa_A^2 - \kappa_B^2\right) + \tfrac{1}{2}\left(\beta_B^2 - \beta_A^2\right). \tag{37}$$

The factor $\frac{1}{2}$ appears here as the first-order Taylor coefficient of the inverse-phase scaling.

Hamiltonian

The standard Newtonian Hamiltonian is obtained by the first ontological collapse. By the Relational Reciprocity Theorem (**??**), an observer self-centers at their own relational origin, meaning observer A measures their own state as $\kappa_A = 0$ and $\beta_A = 0$. (Legacy physics interprets this self-centered zero-state as placing the reference frame at "infinity").

Applying this self-centered zero-state to the linearized two-point law (Eq. 37), the transition collapses into a property of a single state B:

$$E \approx \tfrac{1}{2}\beta_B^2 - \tfrac{1}{2}\kappa_B^2.$$

Multiplying by mc^2 and substituting the legacy definitions of β and κ gives:

$$H = \tfrac{1}{2}mv^2 - \frac{GMm}{r},$$

which is the Newtonian Hamiltonian. The potential information, which belongs to the two-point difference, is reassigned to a single point via an externally postulated potential function $V(r) = -GMm/r$.

Lagrangian

If one starts from the same two-point energy symmetry law $\Delta E_{A\to B} + \Delta E_{B\to A} = 0$ but commits the second ontological collapse — forcing the two distinct points to occupy the same spatial coordinate, $r_A = r_B = r$ — the relational structure collapses further. Setting $\kappa_A^2 = \kappa^2(r)$ and $\beta_A = 0$ while keeping $\beta_B = \beta$, the linearized equation becomes:

$$\Delta E_{A\to B} \approx \tfrac{1}{2}\left(\kappa^2(r) - \kappa^2(r)\right) + \tfrac{1}{2}\beta^2 = \tfrac{1}{2}\beta^2.$$

The potential information completely vanishes from the relational ledger. To recover a dynamical equation, legacy mechanics must separately postulate an external potential energy function $V(r)$ and add it by hand to the kinetic term. Assigning the opposite sign convention to this postulated potential yields the Newtonian Lagrangian:

$$L = \tfrac{1}{2}mv^2 + \frac{GMm}{r}.$$

Both the Hamiltonian and the Lagrangian are revealed as single-point artifacts of collapsing the true two-point phase ratio law.

True Two-point relational form	Collapsed single-point form
$\Delta E_{A\to B} = \left(\frac{\kappa_X}{\beta_Y}\right)_B - \left(\frac{\kappa_X}{\beta_Y}\right)_A$	$L = \frac{1}{2}mv^2 + \frac{GMm}{r}$ $H = \frac{1}{2}mv^2 - \frac{GMm}{r}$
The collapse is performed by Taylor linearizing the phase ratio (introducing the $\frac{1}{2}$), applying the self-centered origin (Hamiltonian), and identifying $r_A = r_B$ (Lagrangian). The two-point phase ratio contains the full energy balance without the need for a separate potential.	

Table 3: True two-point relational law versus collapsed single-point forms.

Summary

Key insight

The Lagrangian and Hamiltonian are linearized artifacts of the single-point collapse of the two-point Energy-Symmetry Law. In the relational framework, all dynamical information is encoded in the dimensionless phase ratio κ_X/β_Y and its geometric closure. No potential function, no force law, and no action principle are required as foundational primitives.

$$\underbrace{\Delta E_{A\to B} + \Delta E_{B\to A} = 0}_{\text{exact symmetry law}} \approx \underbrace{\tfrac{1}{2}\left(\kappa_A^2 - \kappa_B^2\right) + \tfrac{1}{2}\left(\beta_B^2 - \beta_A^2\right)}_{\text{first order approximation}} \approx \underbrace{T \pm V}_{\text{ontological collapse}}$$

Remark 4.4 (Mathematical Status: Groupoid vs. Group)**.** *From a category-theoretic perspective, the relational transitions $\Delta E_{A\to B}$ form a **Groupoid** structure, where operations are defined only between specific connected states. Standard field theories (Hamiltonian/Lagrangian) rely on the collapse of this structure into a **Group** acting on a global manifold (quotienting the state space). Thus, the relational Energy-Symmetry Law operates at the Groupoid level, prior to the reduction into global field theories.*

Legacy Dictionary (for conventional formalisms).
Within RG, all physical content is expressed purely in terms of relational projections β and κ on S^1 and S^2. For readers accustomed to standard frameworks, the following translation rules apply:

1. *General Relativity (metric form):*

$$\kappa_X \mathrel{\hat{=}} \sqrt{-g_{tt}} \quad \text{(static spacetimes)}, \qquad \beta \mathrel{\hat{=}} \frac{\|u^\mu_{\text{spatial}}\|}{u^t c}.$$

2. *Canonical mechanics (Lagrangian/Hamiltonian):* Quantities such as $p_i = \partial L/\partial \dot{q}^i$ do not belong to the ontology of RG. They are secondary derived quantities arising only after collapsing the two-point relational structure into a local single-point formalism.

Here the symbol $\hat{=}$ denotes a pragmatic dictionary entry for translation into legacy notation, rather than an ontological identity.

4.2.3 Newton's Third Law

We now demonstrate that Newton's Third Law, like the Lagrangian and Hamiltonian, emerges as a limiting case of RG. It arises as a necessary mathematical consequence of the same ontological collapse — forcing a two-point relational law into a single-point, instantaneous formalism.

Theorem 4.5 (Newton's Third Law as a Single-Point Limit)**.** *The Energy–Symmetry Law ($\Delta E_{A\to B} + \Delta E_{B\to A} = 0$) mathematically necessitates Newton's Third Law ($\vec{F}_{AB} = -\vec{F}_{BA}$) in the classical, non-relativistic limit where the two-point relational energy budget is collapsed into a single-point potential function $U(\vec{r})$.*

Proof. We begin with the Energy–Symmetry Law (Section 3.10.2), the principle of causal balance for state transitions:

$$\Delta E_{A\to B} + \Delta E_{B\to A} = 0.$$

In the classical limit, this two-point relational law is ontologically collapsed into a single-point potential energy function, U. This function is assumed to depend only on the relative positions of the two interacting entities, A and B:

$$U = U(\vec{r}) \quad \text{where} \quad \vec{r} = \vec{r}_B - \vec{r}_A.$$

This $U(\vec{r})$ is the classical approximation of the system's relational energy ledger. In this collapsed formalism, the force $\vec{F}$ is defined as the negative gradient of this potential.

(1) Force on B by A ($\vec{F}_{AB}$): This force is found by taking the gradient with respect to B's coordinates:

$$\begin{aligned} \vec{F}_{AB} &= -\nabla_B U(\vec{r}_B - \vec{r}_A) && (38) \\ &= -\frac{dU}{d\vec{r}} \cdot \frac{\partial \vec{r}}{\partial \vec{r}_B} && (39) \\ &= -\nabla U(\vec{r}) \cdot \mathbf{I} && (40) \\ &= -\nabla U(\vec{r}) && (41) \end{aligned}$$

(2) Force on A by B ($\vec{F}_{BA}$): This force is found by taking the gradient with respect to A's coordinates:

$$\begin{aligned} \vec{F}_{BA} &= -\nabla_A U(\vec{r}_B - \vec{r}_A) && (42) \\ &= -\frac{dU}{d\vec{r}} \cdot \frac{\partial \vec{r}}{\partial \vec{r}_A} && (43) \\ &= -\nabla U(\vec{r}) \cdot (-\mathbf{I}) && (44) \\ &= +\nabla U(\vec{r}) && (45) \end{aligned}$$

(3) Conclusion: By direct comparison of the results, we find:

$$\vec{F}_{AB} = -\nabla U(\vec{r}) \quad \text{and} \quad \vec{F}_{BA} = +\nabla U(\vec{r}).$$

Therefore, within the single-point collapsed formalism:

$$\boxed{\vec{F}_{AB} = -\vec{F}_{BA}}$$

This completes the proof. Since all physical observations are relational (2.2), embedding the energy budget within an absolute spatial container ($\vec{r}_A, \vec{r}_B$) introduces unobservable parameters, violating Epistemic Hygiene (2.1). To remain empirically grounded, the collapsed potential U must depend exclusively on the measurable physical relation $\vec{r}_B - \vec{r}_A$. Consequently, the law of equal and opposite forces emerges as the mathematical signature of any causally bound, background-independent system $\Delta E_{A\to B} + \Delta E_{B\to A} = 0$. □

4.3 Relational Orbital Mechanics (R.O.M.)

Relational Orbital Mechanics (R.O.M.)

The algebraic derivations and empirical applications of Relational Geometry are isolated in the supplementary document: **R.O.M. paper**. It functions as the operational extension of this document, containing the primary computational tests and phase-resolved orbital mechanics.

This closed algebraic system recovers the leading contributions to perihelion precession, gravitational and transverse Doppler shifts, light deflection, and frame-dragging through elementary closed-form expressions.

High-precision validation shows that all 66 multi-form identities are internally consistent to 50 significant digits and that the leading terms match the corresponding post-Newtonian expansions of general relativity. Higher-order differences remain below the sensitivity of current observations, including those of the S2 star.

Complete phase-resolved orbits and system scales are recovered directly from spectroscopic and chronometric observables, with G and M entering only as unit-conversion factors. Key features of Kerr geometry follow algebraically from the same closure relation. Gravitational waves and the radiative two-body problem remain to be formulated within the relational ontology.

Within R.O.M., we reveal the relational origin of:

- Kepler's Third Law
- Eccentricity
- Vis-Viva
- Classical Acceleration
- Angular Momentum Conservation
- Time-Density Invariant
- Orbital Precession as Phase Accumulation
- Dynamic Event Horizon
- Gravitational Deflection and Lensing

- Schwarzschild Metric
- Kerr Metric
- Algebraic Determination of System Scale

We tested this closed algebraical system on:

- GPS-Earth time dilation
- Mercury's Precession and Jupiter's orbit without mass or G
- Earth–Sun L1 Point Distance Derivation
- S2 Star (Sgr A*) 24 Years Data Alignment Comparison: RG vs GR
- Lense-Thirring: LAGEOS-1 Satellite (Earth orbit)

Relational Orbital Mechanics demonstrates that the entire phenomenology of motion emerges as an algebraic consequence of the Foundational Methodological Principles. All results constitute an unbroken chain of derivation from first principles, maintaining the established epistemic constraints with zero free parameters.

Colab Notebook R.O.M. Full Tests

ROM_FULL_TEST.ipynb - In this notebook we perform three broad tests of Relational Orbital Mechanics:

- Algebraically test the closed system (internal consistency of every multi-form identity). All 66 expressions (50-digit, random c, G, M, a, e, O).
- R.O.M. vs GR: We compute the Schwarzschild geodesic results by high-precision numerical integration, extract their Taylor coefficients in the SAME small parameter, and compare term-by-term to R.O.M.
- ROM reconstructs a, R_s, r_p, r_a and the whole phase-resolved orbit, and recovers GM (a length^3/time^2 combination, no G) and finally M (kg, G used ONLY here as a units conversion factor).

5 Relational Waves (WORK IN PROGRESS)

5.1 Distance as State Difference

Only direct observables enter as inputs. A distance is read off an instrument that was itself calibrated against a state difference, so the state difference is the observable and the length is its translation. Two frames hold one potential state difference between them, projected as κ^2, and the length we call distance is its inverse, $r = R_s/\kappa^2$.

A binary changes its state difference with every revolution. Every frame sharing a projection with that binary carries the change, and each reads it as a variation in length. The legacy model records this variation and interprets it as a wave crossing a fabric of spacetime.

5.2 The Shared Amplitude

Lemma 5.1 (Shared Amplitude Weighting)**.** *An observer holds one relation with the pair, the potential projection κ^2_{obs}, and a fractional transformation ε of the pair enters the observer weighted by that projection:*

$$\kappa^2_{obs} \longrightarrow \kappa^2_{obs}\left[1 + \kappa^2_{obs}\,\varepsilon\right]. \tag{46}$$

Proof. By the Inverse-Distance Potential Amplitude Theorem the projection is the ledger entry itself. On S^2 the unit radius is the constant universal rate of change, distributed by θ_2 between amplitude κ and internal ordering κ_X under $\kappa^2 + \kappa_X^2 = 1$. The observer's ledger therefore allocates the share κ^2_{obs} to the pair, while κ^2 stays the relation internal to A and B. The observer registers the transformation through the share held, so ε arrives as $\kappa^2_{obs}\varepsilon$. □

5.3 Horizon Symmetry

Horizon Symmetry :

Definition 5.2 (Horizon Symmetry)**.** *For two bodies with causal horizons R_{s1} and R_{s2},*

$$\boxed{\eta = \frac{R_{s1}R_{s2}}{(R_{s1}+R_{s2})^2}}$$

Lemma 5.3 (Saturation of Horizon Symmetry). *4η completes the squared horizon difference to unity:*

$$4\eta + \left(\frac{R_{s1} - R_{s2}}{R_{s1} + R_{s2}}\right)^2 = 1. \tag{47}$$

Proof.

$$\frac{4R_{s1}R_{s2}}{(R_{s1} + R_{s2})^2} + \frac{(R_{s1} - R_{s2})^2}{(R_{s1} + R_{s2})^2} = \frac{R_{s1}^2 + 2R_{s1}R_{s2} + R_{s2}^2}{(R_{s1} + R_{s2})^2} = 1. \tag{48}$$

Therefore $4\eta = 1$ at $R_{s1} = R_{s2}$ and $4\eta \to 0$ as $R_{s1} \to 0$. □

4η is the normalised interaction intensity, saturated at identical horizons and empty at a test body, in the same Pythagorean form as $\beta^2 + \beta_Y^2 = 1$ and $\kappa^2 + \kappa_X^2 = 1$.

5.4 The Quadrupolar Argument

Lemma 5.4 (Four Balance Points). *A binary of comparable horizons crosses its mean potential four times per revolution, and the source argument is 2θ.*

Proof. By R.O.M. each active horizon carries two balance points per revolution, where $a = r$ and $\kappa_o^2 = 2\beta_o^2$ hold together:

$$B_a = \arccos(-e), \qquad B_d = 2\pi - \arccos(-e). \tag{49}$$

One active horizon gives two crossings of the mean, so κ_o completes one cycle per revolution. Two active horizons give four crossings, so κ_o completes two cycles per revolution. The source argument counts completed cycles per unit orbital phase, giving 2θ. □

The factor of two counts how many times the state of the pair returns to its own mean in one revolution, fixed by how many horizons are active. This is the relational origin of the quadrupole.

5.5 Source Mismatch

Proposition 5.5 (Source Mismatch). *The binary departs from its per-orbit mean projection by the fraction*

$$\varepsilon(\theta) = 4\eta\, \beta_{src}^2 \cos(2\theta), \qquad \kappa_o^2(\theta) = \kappa^2 \left[1 + \varepsilon(\theta)\right], \tag{50}$$

with κ^2 the mean projection between A and B and β_{src} the orbital projection of the pair.

Proof. By closure $\kappa^2 = 2\beta^2$, the kinematic intensity β_{src}^2 carries the potential mismatch. The horizons share that intensity in proportion to 4η, and the argument is 2θ. Every quantity here belongs to the pair. □

5.6 The Relational Wave

Strain :

Definition 5.6 (Strain). *Strain is the fractional variation of the observer's share of the relation, read as length,*

$$h \equiv \frac{\kappa_{obs}^2}{\kappa_{obs}^2(\theta)} - 1.$$

Theorem 5.7 (Unitless Relational Wave). *For a binary of horizon symmetry η and orbital projection β_{src}, held at shared projection κ_{obs}^2,*

$$\boxed{h(\theta) = \frac{-\,4\eta\, \kappa_{obs}^2\, \beta_{src}^2 \cos(2\theta)}{1 + 4\eta\, \kappa_{obs}^2\, \beta_{src}^2 \cos(2\theta)}} \tag{51}$$

Proof. Write $\mathcal{X}(\theta) = \kappa_{obs}^2 \varepsilon(\theta) = 4\eta\, \kappa_{obs}^2\, \beta_{src}^2 \cos(2\theta)$ for the weighted mismatch, so that $\kappa_{obs}^2(\theta) = \kappa_{obs}^2 \left[1 + \mathcal{X}(\theta)\right]$. Then

$$h(\theta) = \frac{\kappa_{obs}^2}{\kappa_{obs}^2 \left[1 + \mathcal{X}\right]} - 1 = \frac{1}{1 + \mathcal{X}} - 1 = \frac{-\mathcal{X}}{1 + \mathcal{X}}. \tag{52}$$

□

Remark 5.8 (Weak relation). *For $|\mathcal{X}| \ll 1$,*

$$h(\theta) \approx -4\eta\, \kappa_{obs}^2\, \beta_{src}^2 \cos(2\theta), \tag{53}$$

and the amplitude follows κ_{obs}^2, which the observer reads as the $1/D$ falloff.

Remark 5.9 (Bound)**.** *The limits $4\eta \le 1$, $\kappa^2_{obs} \le 1$ and $\beta^2_{src} \le \frac{1}{2}$ give*

$$|\mathcal{X}| \le \tfrac{1}{2}, \qquad h \in \left[-\tfrac{1}{3}, 1\right]. \tag{54}$$

The relational wave stays finite at every state the framework admits.

Remark 5.10 (Correspondence)**.** *Reading κ^2_{obs} in legacy vocabulary as R_s/D with $R_s = 2GM/c^2$, and η as μ/M, the weak relation gives*

$$h \approx \frac{8G\mu\,\beta^2_{src}}{c^2 D}, \tag{55}$$

twice the quadrupole amplitude $4G\mu\beta^2/c^2D$. The historical definition $R_s = 2GM/c^2$ carries that factor of two. Angular and distance structure agree term by term.

Summary

Two frames hold one state difference, projected as κ^2, and read as length. A binary varies that difference, every frame sharing a projection with it carries the variation, and strain is the fractional variation of the share held.

A gravitational wave is the wobble of a relation.

5.7 Density, Mass, and Pressure

5.7.1 Derivation of Density

Translating RG (2DOF) to Conventional Density (3D). In conventional physics, the source term is volumetric density ρ, a 3D concept forced by the "container assumption" — the same ***bi-variable*** syntax: $\underbrace{\text{fixed manifold + metric}}_{\text{structure}} + \underbrace{\text{fields + constants}}_{\text{dynamics}}$ which we explicitly reject with the **Unifying Ontological Principle:** $\boxed{\textbf{SPACETIME} \equiv \textbf{ENERGY}}$ (2.13).

We need to bridge 2DOF S^2- based projection κ with Newtonian "cannonball-like" model of mass-per-volume. We must create a "translation interface" by adopting the conventional units of density, $\rho \propto M/r^3$, as our "translation target".

As shown in **Inverse-Square Theorem** (3.11) **Distance is not a foundational primitive** but relational tension between energy states:

$$r \propto \tfrac{1}{\kappa^2} \text{ when historical units applied and rearranged to solve for } \kappa^2 \text{ this maps to } \tfrac{R_s}{r} = \kappa^2,$$

where κ is the primitive energy projection on the area of unit sphere S^2 relational carrier, and in legacy units $R_s = 2GM/c^2$ links to the mass scale factor $M = E_0/c^2$.

This leads to mass definition:

$$M = \frac{\kappa^2 c^2 r}{2G}$$

To translate this into a volumetric density, we first adopt the conventional 3D (volumetric) proxy, r^3. This is not a postulate of RG, but the first step in applying the legacy (3D) definition of density:

$$\frac{M}{r^3} = \frac{\kappa^2 c^2}{2Gr^2}$$

This expression, however, is incomplete. Our κ^2 "lives" on the 2D surface S^2 (which corresponds to 4π), while the r^3 proxy assumes a 3D volume (which corresponds to $4\pi/3$). For correct normalisation we divide κ^2 by the area of its carrier $1/4\pi$ (we explicitly reject the Newtonian volume integration $(4\pi/3)$ in favor of the S^2 carrier's geometric surface. This enforces that structural ledger scales via the 2DOF S^2 boundary, not by filling an assumed 3D volumetric container).

$$\rho = \frac{1}{4\pi}\left(\frac{\kappa^2 c^2}{2Gr^2}\right)$$

$$\rho = \frac{\kappa^2 c^2}{8\pi G r^2}$$

Local Density $\propto$ Relational Potential Projection κ^2

Maximal Density. At $\kappa^2 = 1$ (the horizon condition (for non rotating systems), $r = R_s$), this density reaches its natural bound, $\rho_{\max}$:

$$\rho_{\max} = \frac{c^2}{8\pi G r^2}$$

Normalized Relation. Thus, our "translation" reveals an identity: the potential projection κ^2 is the ratio of density to the maximal density:

$$\boxed{\kappa^2 = \frac{\rho}{\rho_{\max}} \quad \to \quad \kappa^2 \equiv \Omega}$$

Self-Consistency Requirement

The mass scale factor can be expressed in two equivalent ways.

From the geometric definition:

$$M = \frac{\kappa^2 c^2 r}{2G}.$$

From the energy density:

$$M = \alpha\, r^n\, \rho.$$

Substituting $\rho = \frac{\kappa^2 c^2}{8\pi G r^2}$ into $M = \alpha\, r^n\, \rho$ gives

$$M = \frac{\alpha \kappa^2 c^2 r^{n-2}}{8\pi G}.$$

Equating the two forms:

$$\frac{\alpha r^{n-2}}{8\pi} = \frac{r}{2}.$$

For the mass M to remain a constant independent of the measurement scale r, the exponent must be $n = 3$, yielding $\alpha = 4\pi$. Hence,

$$M = 4\pi r^3 \rho,$$

which closes the consistency loop between the geometric and density-based formulations.

Desmos Project

Derivation of Density

5.7.2 Pressure as Surface Curvature Gradient

In the RG framework pressure is not a thermodynamic assumption but the direct consequence of curvature gradients. The radial balance relation gives

$$P(r) \;=\; \frac{c^4}{8\pi G}\,\frac{1}{r}\,\frac{d\kappa^2}{dr}.$$

Using $\kappa^2 = R_s/r$, one finds $d\kappa^2/dr = -\kappa^2/r$, hence

$$P(r) \;=\; -\,\frac{\kappa^2 c^4}{8\pi G r^2}.$$

Since the local energy density is

$$\rho(r) \;=\; \frac{\kappa^2 c^2}{8\pi G r^2},$$

this yields the invariant equation of state

$$\boxed{P(r) = -\,\rho(r)\, c^2}.$$

Interpretation. P is a surface-like negative pressure (isotropic tension), not a bulk volume pressure. It expresses the resistance of energy-geometry itself to changes in projection.

Consistency. If one formally freezes the projection parameter ($d\kappa^2/dr = 0$), then $P = 0$. But in this case the angular curvature terms remain uncompensated, and the field equation is no longer satisfied. Any nontrivial radial dependence of κ inevitably generates the negative tension

$$P = -\rho c^2,$$

which precisely cancels the residual curvature. Thus the negative pressure is not optional but a necessary ingredient for full self-consistency.

Desmos Project

Derivation of Density and Pressure

Maximum pressure. At the geometric bound $\kappa^2 = 1$ (horizon condition), the density saturates at

$$\rho_{\max} = \frac{c^2}{8\pi G r^2},$$

and the corresponding pressure is

$$P_{\max} = -\rho_{\max} c^2 = -\frac{c^4}{8\pi G r^2}.$$

This negative surface pressure represents the ultimate tension limit of spacetime fabric at a given scale r.

Pressure in WILL is the intrinsic surface tension of energy-geometry, saturating at $P_{\max} = -c^4/(8\pi G r^2)$.

Physical meaning

The negative pressure is not an exotic substance but the geometric tension required to maintain self-consistency when κ^2 varies radially. It's the relational analogue of surface tension in a soap bubble.

5.8 Unified Relational Field Equation

From the energy-geometry equivalence, the complete description of gravitational phenomena reduces to a single algebraic relation linking the geometric scale to the energy density ratio:

$$\boxed{\kappa^2 = \frac{R_s}{r} = \frac{\rho_{field}}{\rho_{max}}}$$

This identity defines the **local energy state of the relational geometry itself**. Here $\rho_{\max} = c^2/(8\pi G r^2)$ is the saturation density limit, and ρ_{field} is the effective energy density of the relational curvature.

Field Equation and Matter Sources

For a static, spherically symmetric configuration containing matter with density $\rho_{\text{matter}}(r)$, the relationship is governed by the differential accumulation of the potential:

$$\boxed{\frac{d}{dr}(r\,\kappa^2) = \frac{8\pi G}{c^2}\,r^2 \rho_{matter}(r)} \tag{56}$$

Remark 5.11 (Scope of the reproduced sector)**.** *The differential form above reproduces the static tt-component. This is not the boundary of the framework: through phase parametrization the same projections describe any single relational orbit — bound at any eccentricity $e = 2\beta_p^2/\kappa_p^2 - 1$, and unbound (deflection, lensing)– and rotating systems (R.O.M. Kerr without metric). GR direct parameters translational mapping. Current boundaries: sec:challenges.*

The Vacuum Solution ($\rho_{matter} = 0$). In the vacuum region outside a central mass, the source density vanishes ($\rho_{matter} = 0$). The field equation implies conservation of the projection budget:

$$\frac{d}{dr}(r\kappa^2) = 0 \implies r\kappa^2 = \text{const} = R_s.$$

Thus, we recover the potential law of WILL RG:

$$\boxed{\kappa^2 = \frac{R_s}{r}.}$$

Resolution of Roles

1. **The Identity** $\kappa^2 = \rho/\rho_{\max}$ describes the state of the *field* geometry.
2. **The Equation** $(r\kappa^2)' \sim \rho_{\text{matter}}$ describes how *matter* generates geometry.

In vacuum, the generator is zero, but the field persists as the algebraic structure $\kappa^2 = R_s/r$.

5.9 No Singularities Within Admissible Relational Range

WILL Relational Geometry resolves the singularity problem by naturally constraining the valid range of radial distance. Minimal radial distance is kept at $r_{min} = R_s/2$ by the fundamental invariance of the light-speed limit. Parameters stay finite, and energy remains bounded. Black holes become the boundary of relational difference between energy states.

Distance is not a foundational primitive but relational tension between energy states (3.11) The amplitude bounds $\kappa^2 \le 1$ (static case) and $\kappa^2 \le 2$ (extremal rotation) cap the radial distance by the closure theorem (3.6):

No Singularities

$$r = R_s/\kappa^2 \geq R_s/2$$

$$\beta_{max}^2 = 1 \Longrightarrow \kappa_{max}^2 = 2 \Longrightarrow r_{min} = R_s/\kappa_{max}^2 \Longrightarrow r_{min} = R_s/2 \tag{57}$$

The point $r = 0$ is absent from the admissible relational range.
The same bound enforces:

$$\rho \leq 2\rho_{\max} \quad \text{and} \quad |P| \leq |2P_{\max}| = c^4/(8\pi G r^2) \tag{58}$$

Another interesting hypothetical explanation of this phenomenon comes from Relational Orbital Mechanics. The Dynamic Relational Horizon provides a new point of view on the nature of Black Holes, where precession of the perihelion becomes equivalent to the orbital frequency $\Delta\varphi = 2\pi$. The orbit starts to fold onto itself and nullifies any orbital motion. It is not a two-surface enclosing a region — so there is no interior to be hidden in the first place. However, the concept of the dynamic horizon remains purely mathematical and requires more research.

Derivation Chain Map

Methodological constraints:

↓(*Epistemic Hygiene+Relational Origin+Ontological Minimalism+Math Transparency*)↓

Relational Closure + Causal Continuity + Isotropy Theorems

↓ *(apply to existing physics)* ↓

Standard Primitives: $\underbrace{\text{background manifold + metric}}_{\text{spacetime structure (not derived)}} \quad + \quad \underbrace{\text{G, c, M, fields}}_{\text{energy dynamics (postulated)}}$

↓ *(Ontological reduction by Relational Origin Principle — no background allowed)* ↓

One Primitive: ***WILL*** ≡ ***SPACE-TIME-ENERGY*** $\Longrightarrow$ $\boxed{\text{SPACETIME} \equiv \text{ENERGY}}$

↓ *(by Relational Closure + Causal Continuity + Isotropy: derive relational energy carriers)* ↓

S^1 **(1-DOF kinematic) +** S^2 **(2-DOF potential)**

↓ *(by Duality of Relation Lemma)* ↓

Relational Conservation Theorem: $\underbrace{\text{Amplitude}^2}_{\text{External Transformation}} + \underbrace{\text{Phase}^2}_{\text{Internal Change}} = 1$

$\boxed{\underbrace{1/\beta_Y = 1 + z_\beta}_{\text{transverse Doppler shift}} \quad \underbrace{\beta^2 + \beta_Y^2 = 1}_{\text{Kinematic } S^1}}$ **Relational Projections** $\boxed{\underbrace{\kappa^2 + \kappa_X^2 = 1}_{\text{Potential } S^2} \quad \underbrace{1/\kappa_X = 1 + z_\kappa}_{\text{gravitational redshift}}}$

↓ *(by DOF-Indifference Lemma: equal quadratic weight to each independent DOF)* ↓

Closure Theorem: $\underbrace{1\,\text{Amplitude}^2}_{\text{on } S^2} = \underbrace{2\,\text{Amplitude}^2}_{\text{on } S^1} \Longrightarrow \boxed{\kappa^2 = 2\beta^2}$

↓ *(by Relational Conservation Theorem derive metric intervals as scaffolding)* ↓

Minkowski: $\underbrace{\beta^2 + \beta_Y^2 = 1}_{\text{irreducible primitive}} + \underbrace{x, y, z, t...}_{\text{coordinate inflation}} = \underbrace{c^2 d\tau^2 = c^2 dt^2 - dx^2 - dy^2 - dz^2}_{\text{mathematical scaffolding}}$

Schwarzschild: $\underbrace{\kappa^2 + \kappa_X^2 = 1}_{\text{irreducible primitive}} + \underbrace{x, y, z, t...}_{\text{coordinate inflation}} = \underbrace{c^2 d\tau^2 = c^2 (1 - (R_s/r))\, dt^2}_{\text{mathematical scaffolding}}$

↓ *(by Relational Closure and Causal Continuity Theorems: any change must be balanced)* ↓

Energy Symmetry Law: $\boxed{\Delta E_{A\to B} + \Delta E_{B\to A} = 0}$

↓ $\Delta E_{A\to B} = E_0(\kappa_{X,B}/\beta_{Y,B} - \kappa_{X,A}/\beta_{Y,A})$; $\Delta E_{B\to A} = E_0(\kappa_{X,A}/\beta_{Y,A} - \kappa_{X,B}/\beta_{Y,B})$ ↓

Speed of light: $\beta \to 1 \;\to\; \beta_Y \to 0 \;\to\; E = E_0 \cdot \kappa_X/0 \to \infty \;\to$ **requires** ∞ **energy**

Event Horizon: $\kappa \to 1 \;\to\; \kappa_X \to 0 \;\to\; E = E_0 \cdot 0/\beta_Y \to 0 \;\to$ **requires** 0 **energy**

↓ *(by Unified Relational Scaling Lemma)* ↓

Equivalence Principle $\boxed{m_g \equiv m_i \equiv m = E_0/c^2}$

↓ *(**Lagrangian and Hamiltonian** derived as collapsed single-point approximations)* ↓

$\underbrace{\Delta E_{A\to B} + \Delta E_{B\to A} = 0}_{\text{exact symmetry law}} \approx \underbrace{\tfrac{1}{2}(\kappa_A^2 - \kappa_B^2) + \tfrac{1}{2}(\beta_B^2 - \beta_A^2)}_{\text{first order approximation}} \approx \underbrace{T \pm V}_{\text{ontological collapse}}$

↓ *(apply RG to orbital mechanics)* ↓

Relational Orbital Mechanics $\boxed{\textbf{R.O.M.}}$**: a closed algebraic system of equations**

↓*(by Factorization Theorem: every quantity X splits into unitless core $\hat{X}$ and scale label S_X)*↓

Unitless Core: $\underbrace{X = \hat{X}(,e)}_{\text{unitless core}}\underbrace{S_X(R_s, c, G)}_{\text{units-scale}}$ **G and M are unit-conversion factors.**

↓ *(by Inverse Square, Closure and Causal Continuity Theorems)* ↓

No Singularities: $\beta_{\max}^2 = 1 \;\to\; \kappa_{\max}^2 = 2 \;\to\; r_{\min} = R_s/\kappa_{\max}^2 \;\to\; r_{\min} = R_s/2$

The point $r = 0$ **is absent from the admissible relational range**

↓ *(apply 2D to 3D translation interfaces to derive Relational Field Equation:)* ↓

$\boxed{\kappa^2 = R_s/r = \rho_{\text{field}}/\rho_{\max}} \Longrightarrow \boxed{\text{Spacetime Geometry}(R_s/r)} \equiv \boxed{\text{Energy Density}(\rho/\rho_{\max})}$

↓ **Theoretical Ouroboros** ↓

$\boxed{\text{SPACETIME} \equiv \text{ENERGY}} \Leftrightarrow \boxed{\text{SPACETIME GEOMETRY} \equiv \text{ENERGY DENSITY}}$

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6 Stage IV Consequences and Direct Applications

6.1 Theoretical Ouroboros

Closure

Ontological principle is proven as the inevitable consequence of geometric consistency. Field Equation $\Longleftrightarrow$ Ontological Principle

We have shown that Unifying Ontological Principle (2.13), through pure geometric reasoning, necessarily leads to an equation which mathematically expresses the very same equivalence we began with. We started with SPACETIME $\equiv$ ENERGY, from which geometry and physical laws are logically derived, and these derived laws then loop back to intrinsically define and limit the very nature of energy and spacetime, proving the self-consistency of the initial idea.

$$\boxed{\text{SPACETIME} \equiv \text{ENERGY.}}$$

The ratio of geometric scales equals the ratio of energy densities.

$$\boxed{\kappa^2 = \frac{R_s}{r} = \frac{\rho}{\rho_{max}}}$$

$$\boxed{\begin{array}{c}\text{SPACETIME GEOMETRY}\\ \equiv \\ \text{ENERGY DISTRIBUTION}\end{array}}$$

Summary

All physical structure emerges from the single relational equivalence:

$$\boxed{\text{SPACETIME} \equiv \text{ENERGY}}$$

From this, by enforcing geometric self-consistency, one necessarily arrives at the Unified Geometric Field Equation:

$$\boxed{\kappa^2 = \frac{R_s}{r} = \frac{\rho}{\rho_{\max}}.}$$

This is not an external law but an intrinsic closure relation: geometry and energy are two mutually defining projections of a single entity. It represents the completion of the theoretical Ouroboros — where the principle generates its own mathematical expression and the expression in turn validates the principle.

From a philosophical and epistemological point of view, this can be considered the crown achievement of any theoretical framework - the "Theoretical Ouroboros". But regardless of the aesthetic beauty of this result, let us remain sceptical.

6.2 $W_{\mathbf{ILL}}$: Unity of Relational Structure

The ontological principle

$$\text{SPACETIME} \equiv \text{ENERGY}$$

states that there is only one closed relational ledger - WILL. What we call space, time and energy are different projections of the same relational structure. For any energy-closed system observed from a relational origin, this resource appears through four operational projections:

Desmos project

WILL= 1 Derivation of the W_{ILL} invariant.

These quantities are defined as correlated projections of the same underlying WILL structure. In the dimensionful form we write:

$$M = \frac{\beta^2}{\beta_Y}\frac{c^2 a}{G}, \qquad E = \frac{\kappa^2}{\kappa_X}\frac{c^4 a}{2G}, \qquad T = \kappa_X(\frac{2Gm_0}{\kappa^2 c^3})^2, \qquad L = \beta_Y(\frac{Gm_0}{\beta^2 c^2})^2,$$

where (β, β_Y) and (κ, κ_X) are the kinematic and gravitational projections on the carriers S^1 and S^2, m_0 is the rest mass, $E_0 = m_0 c^2$ is the rest energy, and a is the relational scale of the system as semi-major axis (average length per cycle within this system).

The relational conservation of the carriers $\beta^2 + \beta_Y^2 = 1, \quad \kappa_X^2 + \kappa^2 = 1$, together with the closure theorem $\kappa^2 = 2\beta^2$, fix a unique dimensionless combination of these four projections. Combining E, T, M, and L we obtain

$$W_{\text{ILL}} \equiv \frac{ET}{ML} = \frac{\dfrac{E_0}{\kappa_X}\,\kappa_X t^2}{\dfrac{m_0}{\beta_Y}\,\beta_Y a^2} = \frac{E_0 t^2}{m_0 a^2}.$$

By the relations that tie temporal $t = a/c$ and spatial $a = R_s/\kappa^2$ scales this ratio is identically equal to unity for any relationally closed system:

$$\boxed{W_{\text{ILL}} = \frac{ET}{ML} = 1.}$$

All dimensionful constants cancel automatically; the value is fixed by the geometry of the carriers, not by a choice of units.

Desmos project

WILL EARTH GPS calculates GPS satellite time shift and testing $W_{ILL} = 1$ on the Earth-GPS system

The same invariant can be written in a phase–normalized form, using local projections

$$E_o = \frac{E_0}{\kappa_{Xo}}, \qquad M_o = \frac{m_0}{\beta_{Yo}}, \qquad T_o = \kappa_{Xo} t_o^2, \qquad L_o = \beta_{Yo} r_o^2,$$

Equivalently, for any state (β_o, κ_o), any scale r_o and any phase along the orbit. Then

$$\boxed{W_{\text{ILL}} = \frac{E_o T_o}{M_o L_o} = 1 \quad \text{for ALL ENERGIES, ALL SCALES and ALL PHASES.}}$$

$$\frac{E_o}{M_o} = \frac{L_o}{T_o},$$

so the energy sector and the spacetime sector are not independent. Every change of the relational state rescales (E_o, M_o) and (T_o, L_o) coherently so that this equality is always preserved. The familiar practice of treating energy–mass and space–time as separate blocks is therefore an interpretational approximation. This equation does not bring any new predictive power, but its ontological meaning shows that any coherent unit system will naturally collapse into unity, dissolving the false separation:

$$\text{SPACETIME} \equiv \text{ENERGY}$$

$$\boxed{W_{ILL} = 1}$$

One conserved relational resource appears as mass, energy, time and length, but always in a way that keeps their ratio fixed.

Interpretive Note: The Name "WILL"

The term **WILL** stands for **SPACE-TIME-ENERGY**. It is both a formal shorthand and a philosophical statement: the universe is not a stage where energy acts through time upon space, but a single self-balancing structure whose internal distinctions generate all phenomena. The name also serves as a gentle irony toward anthropic thinking: the Cosmos does not possess "will" - yet through WILL, it manifests All that Is.

Summary

$$\boxed{\textbf{WILL} \equiv \frac{ET}{ML} = 1 \quad \Longleftrightarrow \quad \text{Geometry} = \text{Energy} = \text{Causality.}}$$

WILL is not the unit of something - but the Unity of Everything.

6.3 Earth–GPS System: Standard GR 1PN as First-Order Approximation of RG

We have made many derivations and daring claims. It's time to put them to the test. For our first tests we will use the standard Earth - GPS satellite system to challenge theorems and lemmas: (2.5; 2.7; 3.3; 3.6; 3.7; 3.20; 4.2; 3.22; 3.15; 3.10.2; 6.2; 4.2.2; 4.2.3)

Statement of the Verification Problem

The standard post-Newtonian expression for the secular time shift between a clock on the Earth's surface and a clock on a circular GPS orbit, accumulated over one solar day, is

$$\Delta t_{\mathrm{GR}} \;=\; \left(\frac{\Phi_{\mathrm{GPS}}-\Phi_E}{c^2} \;-\; \frac{v_{\mathrm{GPS}}^2-v_E^2}{2\,c^2}\right) D\,M, \tag{59}$$

with $D = 86{,}400\,\mathrm{s}$, $M = 10^6$, $\Phi(r) = -GM_\oplus/r$, $v_E = 2\pi R_E/D$, and $v_{\mathrm{GPS}} = \sqrt{GM_\oplus/r_{\mathrm{GPS}}}$.

WILL Relational Geometry (WILL RG) expresses the same time shift as an exact algebraic ratio of two-carrier projection invariants $\tau \equiv \sqrt{1-\kappa^2}\,\sqrt{1-\beta^2}$:

$$\Delta t_{\mathrm{RG}} \;=\; \left(1 \;-\; \frac{\sqrt{1-\kappa_E^2}\,\sqrt{1-\beta_E^2}}{\sqrt{1-\kappa_{\mathrm{GPS}}^2}\,\sqrt{1-\beta_{\mathrm{GPS}}^2}}\right) D\,M, \tag{60}$$

where $\kappa^2(r) = 2GM_\oplus/(rc^2)$, $\beta^2 = v^2/c^2$, and the orbital closure condition $\beta_{\mathrm{GPS}}^2 = \kappa_{\mathrm{GPS}}^2/2$ holds for circular motion.

The question addressed in this section is whether (59) is the leading-order Taylor expansion of (60) in the small parameters κ^2 and β^2. If so, the algebraic difference of the two coefficients must vanish identically at first order, and the residual at finite values of the small parameters must reproduce the analytically derived second-order correction.

Methodology

The verification is split into two independent parts, both executed in a single public notebook (Section 6.3).

Part I — symbolic identity. Introduce the dimensionless geometric ratio $\rho \equiv R_E/r_{\mathrm{GPS}}$, which is treated as a finite (order-unity) parameter and is *not* expanded. The exact WILL RG shift coefficient is then

$$\delta_{\mathrm{RG}}(\kappa_E^2, \beta_E^2, \rho) \;=\; 1 - \frac{\sqrt{(1-\kappa_E^2)(1-\beta_E^2)}}{\sqrt{(1-\kappa_E^2\rho)(1-\kappa_E^2\rho/2)}}. \tag{61}$$

A single bookkeeping parameter ε is introduced via $\kappa_E^2 \to \varepsilon\,\kappa_E^2$, $\beta_E^2 \to \varepsilon\,\beta_E^2$, and the Taylor series of (61) is computed to second order in ε using `sympy.series`. The first-order coefficient $\delta_{\mathrm{RG}}^{(1)}$ is then subtracted from the dimensionless form of the GR coefficient δ_{GR} obtained from (59) under the substitutions $\Phi/c^2 = -\kappa^2/2$ and $v^2/c^2 = \beta^2$. The result is simplified algebraically.

Part II — numerical comparison. Both (59) and (60) are evaluated with 40-digit precision arithmetic (`mpmath`, `mp.dps = 40`) using a single set of input parameters: $c = 299{,}792{,}458\,\mathrm{m\,s^{-1}}$, $GM_\oplus = 3.986004418 \times 10^{14}\,\mathrm{m^3\,s^{-2}}$, $R_E = 6{,}378{,}137\,\mathrm{m}$, $r_{\mathrm{GPS}} = 26{,}561{,}750\,\mathrm{m}$, $D = 86{,}400\,\mathrm{s}$. The closure residual $\beta_{\mathrm{GPS}}^2 - \kappa_{\mathrm{GPS}}^2/2$ is monitored as a numerical consistency check. The numerical discrepancy $\Delta t_{\mathrm{RG}} - \Delta t_{\mathrm{GR}}$ is then compared with the analytically predicted second-order correction $\delta_{\mathrm{RG}}^{(2)}\,D\,M$ obtained from Part I.

Results

Symbolic. The first-order Taylor coefficient of the WILL RG shift, expressed in κ_E^2, β_E^2, and the geometric ratio ρ, is

$$\delta_{\mathrm{RG}}^{(1)} \;=\; \frac{\kappa_E^2}{2} \;+\; \frac{\beta_E^2}{2} \;-\; \frac{3\,\kappa_E^2\,\rho}{4}. \tag{62}$$

The GR coefficient, expressed in the same variables, is

$$\delta_{\mathrm{GR}} \;=\; \frac{\kappa_E^2}{2} \;+\; \frac{\beta_E^2}{2} \;-\; \frac{3\,\kappa_E^2\,\rho}{4}. \tag{63}$$

The algebraic difference $\delta_{\mathrm{RG}}^{(1)} - \delta_{\mathrm{GR}}$ vanishes identically.

The second-order content of (61), which is absent from the additive GR formula (59), is

$$\delta_{\mathrm{RG}}^{(2)} \;=\; \frac{\kappa_E^4}{8} \;+\; \frac{\beta_E^4}{8} \;-\; \frac{\beta_E^2\,\kappa_E^2}{4} \;+\; \frac{3\,\kappa_E^4\,\rho}{8} \;-\; \frac{19\,\kappa_E^4\,\rho^2}{32} \;+\; \frac{3\,\beta_E^2\,\kappa_E^2\,\rho}{8}. \tag{64}$$

Three structural features of (64) have no analogue in any rearrangement of the additive GR sum (59): (i) the pure quadratic contributions $\kappa_E^4/8$ and $\beta_E^4/8$ arising from higher-order expansion of $\sqrt{1-x}$; (ii) the cross-term $-\beta_E^2\kappa_E^2/4$, which originates from the multiplicative coupling of the gravitational and kinematic projection factors within $\tau_E/\tau_{\mathrm{GPS}}$; and (iii) the ρ^2 contribution, which encodes nonlinear dependence on the relative depth of the gravitational well.

Numerical. With the parameter set listed above:

quantity	value [μs day^{-1}]
Δt_{RG} (exact)	38.5421472752
Δt_{GR} (first order)	38.5421472451
$\Delta t_{\text{RG}} - \Delta t_{\text{GR}}$	3.017×10^{-8}
$\delta^{(2)}_{\text{RG}} \cdot D\,M$ (predicted)	3.017×10^{-8}
residual	$\sim 10^{-17}$

The numerical discrepancy $\Delta t_{\text{RG}} - \Delta t_{\text{GR}}$ matches the analytically derived second-order correction $\delta^{(2)}_{\text{RG}}\,D\,M$ to within the limit set by the 40-digit floating-point arithmetic. The closure residual $\beta^2_{\text{GPS}} - \kappa^2_{\text{GPS}}/2$ vanishes to $\sim 10^{-50}$. The relative magnitude of the term omitted by the GR additive formula is $\sim 7.8 \times 10^{-10}$ of the total shift.

Interpretation

Equations (62)–(63) establish that the additive post-Newtonian formula (59) coincides exactly with the linear Taylor coefficient of the WILL RG exact algebraic ratio (60). This is an algebraic identity in $(\kappa^2_E, \beta^2_E, \rho)$, not a numerical near-match: the additive sum-of-corrections structure of (59) is recovered as the unique leading-order content of the multiplicative ratio τ_E/τ_{GPS}.

For the Earth–GPS configuration, the second-order term (64) contributes at the level of $\sim 3 \times 10^{-8}\,\mu$s day^{-1}, i.e. $\sim 8 \times 10^{-10}$ of the total accumulated shift. This is below the present precision of operational GPS clock comparisons and does not, by itself, constitute a distinguishing observation between the two formulations in this regime. The structural difference between an additive and a multiplicative composition of projection factors — in particular the presence of the $\beta^2\kappa^2$ cross-term in (64) — becomes quantitatively relevant only when κ^2 or β^2 approach order unity.

The scope of the present statement is restricted to the relation between the WILL RG closed-form expression (60) and the *additive post-Newtonian* form of the GR time-dilation formula (59). A comparison against fully non-linear geodesic integration in the exact Schwarzschild geometry combined with special-relativistic kinematics for the rotating Earth surface is a separate computation and is not addressed in this section.

Reproducibility

The complete symbolic and numerical verification reported in this section is contained in a single public Google Colab notebook, comprising the two scripts used to produce (62), (64), and the numerical table above:

Desmos project

Earth-GPS time, energy symmetry and WILL invariant test

Colab Notebook

Earth-GPS 1st order RG=GR.ipynb

The notebook has no external dependencies beyond `sympy` and `mpmath` and is self-contained. All input parameters are listed explicitly in the numerical script; any change in r_{GPS}, R_E, or $GM_\oplus$ rescales both Δt_{RG} and Δt_{GR} consistently and does not affect the symbolic identity $\delta^{(1)}_{\text{RG}} \equiv \delta_{\text{GR}}$.

6.4 Ontological Shift: From Descriptive to Generative Physics

In conventional physics the methodology follows a descriptive paradigm:

1. Observable phenomena are identified.
2. Empirical regularities are codified as "laws of nature."
3. Mathematical formalisms are constructed to *describe* these regularities.

Thus, physical laws enter as external assumptions that model what is seen. Even in General Relativity, where geometry plays the central role, the equivalence principle and the metric postulate remain external inputs.

RG inverts this paradigm. Laws are generated as inevitable consequences of relational geometry:

- Concepts such as "inertial mass equals gravitational mass" emerge automatically as algebraic identities enforced by the geometry.
- What classical physics calls "laws of nature" serve as secondary shadows of the single relational principle:

$$\text{SPACETIME} \;\equiv\; \text{ENERGY}.$$

Summary

Standard Physics: Laws *describe* what we observe.
Relational Geometry: Laws are *generated* as necessary products of closure and self-consistency.

In this sense, the ontological role of physical law is transformed. Physics ceases to be a catalog of empirical descriptions, and becomes the logical unfolding of a single relational structure. WILL identifies the necessary conditions under which all observed phenomena arise.

Descriptive Physics (Standard)	**Generative Physics (WILL)**
Phenomena are *observed* first, then summarized into empirical laws.	Laws emerge as *inevitable consequences* of relational geometry.
Physical laws are *assumptions* introduced to model reality.	Physical laws are *identities*, enforced by geometric self-consistency.
Time and space are treated as external backgrounds.	Time and space are *projections of energy relations.*
Dynamics = evolution of states *in time.*	Dynamics = ordered succession of balanced configurations; *time is emergent.*
Goal: *describe* what is observed.	Goal: *show why nothing else is possible.*

Table 4: Ontological contrast between standard descriptive physics and the generative paradigm of WILL Relational Geometry.

6.4.1 Methodology comparison of GR with RG

Webpage Link

https://willrg.com/RGvsGR/

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Phenomenon	Standard (GR) Result	Relational Geometry (RG)
GPS time shift / gravitational redshift	Frequency shift = summation of kinetic (SR) and gravitational (GR) effects.	Single symmetric law: $\tau = \beta_Y \cdot \kappa_X$, single multiplicative parameter. 38.52 μs/day verified directly with GPS satellites.
Photon sphere, ISCO, horizons	Derived by solving geodesic equations in Schwarzschild metric.	Critical radii emerge from simple symmetries (Photon sphere: $\theta_1 = \theta_2 = 54.73°$ ("magic angle") $Q^2 = \kappa^2 + \beta^2 = 1$. ISCO: $Q^2 = Q_Y^2 = 1/2$).
Mercury's perihelion precession	Complex expansion of Einstein field equations.	Same number obtained from RG with $\Delta\varphi = \frac{2\pi \cdot \tau_Y^2}{1-e^2} = 43''$/century, using simple algebra.
Cosmological redshift	Photon "loses energy" as universe expands.	Energy conserved; redshift = redistribution of projection parameters. (Details in WILL PART II)
Cosmological absolute scale (Supernovae fit)	Hubble-like expansion, ΛCDM fits	$H_0 \equiv \sqrt{8\pi G \rho_\gamma / (3\alpha^2)} = 68.15$ Derived from CMB temperature and α connecting micro and macro scales (Details in WILL PART II-III).
Cosmological constant Λ	Added by hand to fit data ("dark energy").	Arises naturally as $\Lambda = 2/3r^2$. No extra entities required. (More details in WILL PART I and II)
Singularities	Predicted in black holes and big bang ($\rho \to \infty$).	Structurally absent: The point $r = 0$ is absent from the admissible relational range. Density bounded by $\rho_{\max} = c^2/(8\pi G r^2)$.
Local gravitational energy	Indirectly localized at infinity (ADM/Bondi).	Directly measurable via κ, e.g. from light deflection angle or redshift.
Unification with QM	Absent in standard GR framework.	Same projectional law applies from microscopic $\alpha = \beta_1$ (QM) to cosmic $\kappa^2 = \Omega_\Lambda$ (GR, COSMO) scales. (Details in WILL PART II and III)

Table 5: Classical GR results vs. WILL RG outcomes. Known effects are recovered by simpler symmetric laws, while new predictions eliminate singularities and describe cosmology without additional dark components.

6.5 Discussion

All WILL RG results, including the success of R.O.M., remain a single unbroken generative chain of derivations from methodological constraints with zero flexibility. Relational Orbital Mechanics functions as the direct operational consequence of applying the Foundational Methodological Principles to bound gravitational systems.

Epistemic Benefits

WILL RG enforces epistemic hygiene by operating exclusively on dimensionless, operationally defined observables. By replacing the absolute background manifold with the topological closure of the S^1 and S^2 carriers, the framework eliminates surplus ontology — absolute space, absolute time, and independent mass. Every parameter in the algebraic chain (β, κ, e, T, z) maps directly to a measurable spectroscopic or chronometric quantity. Consequently, the theory achieves maximum ontological parsimony: it posits only what is structurally necessary to close the relational ledger, stripping away the metaphysical artifacts inherited from classical continuum mechanics. The contrast in the number of required ontological primitives is instructive:

Primitives of General Relativity

1. A differentiable manifold (the spacetime background/container).
2. A pseudo-Riemannian metric tensor $g_{\mu\nu}$ defined on that manifold.
3. A stress-energy tensor $T_{\mu\nu}$ representing matter/energy as an independent entity.
4. The gravitational constant G and the speed of light c as fundamental coupling constants.
5. The Einstein Field Equations (a postulated dynamical law linking the geometry to the energy).

6. The Equivalence Principle (a postulated axiom linking inertial and gravitational mass).

Primitives of Relational Geometry

1. Methodological constraints (2.1, 2.2, 2.3. 2.4). *Not statements about the nature of reality, but strict rules of logical and epistemic purity.*

A Note on Methodology vs. Ontology

It may be objected that the foundational principles constitute axioms, thereby contradicting the claim of *zero assumptions*. This objection conflates *methodological constraints* with *ontological axioms*.
Under the strictures of the Scientific Method, the burden of proof rests exclusively on the claimant asserting an absolute existence. Refusing to import unobserved absolutes (such as a background container or intrinsic properties) is not an assumption; it is the default epistemic stance — the null hypothesis.
These principles function as rules of logical engagement to prevent the introduction of surplus ontology. To challenge them is to argue for the admission of a specific unobserved absolute — and under the empirical foundation of physics, the burden of proof rests with that admission.

Operational Benefits: Dispensing with Differential Formalism

The operational consequence of the relational construction is the removal of differential equations of motion and central acceleration from the list of foundational primitives.

In standard orbital mechanics, the central acceleration vector (Newtonian or geodesic) must be postulated and integrated to derive orbital periods, shapes, precession, and equilibrium configurations. In WILL RG, the same observables are obtained from algebraic identities among the projections on the relational carriers S^1 and S^2.

This absence of differential formalism is a consequence of the generative structure. By enforcing topological closure ($\kappa^2 = 2\beta^2$) and maximal symmetry from the outset, the allowed relational states of a bound system are classified by their dimensionless projections. The mathematics requires no integration of trajectories, no effective potentials, and no auxiliary constructs such as rotating reference frames.

Observables that in conventional treatments appear only after lengthy perturbative expansions or numerical integration emerge here as closed-form consequences of the relational geometry. This is demonstrated across all derivations in this paper, including the period–semi-major axis relation, secular precession, light deflection, and the location of the Sun–Earth L1 point.

6.5.1 Predictions, Falsifiability, and Open Research

Complete methodological transparency is the necessary future of scientific inquiry. We adopt and promote the **Open Research** philosophy. All predictions are published alongside the raw data and computational tools, ensuring reproducibility. This philosophy cultivates research as an international open collaboration driven by shared empirical scrutiny rather than theoretical orthodoxy.

The **WILL RG Trilogy** generates parameter-free predictions across 20 orders of magnitude—from atomic structure to the cosmic horizon. The updated list of predictions, computational tools, and derivations is maintained at willrg.com, with the development history accessible via the public repository at github.com/AntonRize/WILL.

A falsifiable test of this framework lies in the high-precision monitoring of the S2 star at the Galactic Centre. Current GRAVITY data are statistically consistent with both RG and GR. However, the algebraic phase accumulation of RG dictates a second-order precessional correction $\mathcal{O}(\beta^4)$ that differs from the 2PN expansion of General Relativity. This higher-order correction contains zero free parameters. A detection at the predicted level, or a sufficiently tight upper limit below it during future pericentre passages, provides an empirical test of the relational ontology against the metric manifold.

6.5.2 Challenges and Current Limitations

Gravitational waves and the radiative two-body problem remain to be formulated within the relational framework. The current construction addresses conservative, bound orbits. Extending it to dissipative, radiative dynamics requires a relational description of propagating degrees of freedom on the carriers and remains an open technical task.

A self-consistent multi-body architecture that extends beyond the collinear L1 case awaits algebraic realisation. The nested-contour approach provides a conceptual route, but its mathematical formulation for systems containing three or more gravitationally bound bodies (including mean-motion resonances and long-term stability) requires further development.

Progress on both fronts is currently limited by mathematical facility rather than by any intrinsic prohibition within the relational ontology. Collaboration with researchers experienced in radiative dynamics and in the algebraic treatment of multi-body closures is invited.

6.6 Conclusion

Relational Geometry demonstrates that the phenomenology of motion — from Keplerian architecture to perihelion precession and light deflection — emerges as an algebraic consequence of the Foundational Methodological Principles.

WILL RG operates as the algebraic core that remains when the ontological superstructure of Riemannian geometry, differential equations, and tensorial formalism is removed. The S2 orbit fit and Mercury's precession demonstrate that RG functions as an empirically viable ontology that reproduces the data with a simpler closed-form relational algebra and zero structural commitments to a background container.

The structural consequences of this framework are:

1. **G and M are no longer required as primitives.** The Schwarzschild length R_s and all orbital distances are derived from dimensionless spectroscopic and chronometric ratios. The Newtonian constant G and the kilogram mass serve as post-hoc conversion factors when translating geometric results into legacy units.

2. **Topological origin of Metrics.** The Minkowski, Schwarzschild, and Kerr intervals are revealed as coordinate-inflated Pythagorean identities of the relational carriers S^1 and S^2. Relativistic effects (time dilation and length contraction) emerge as geometric phase rotations.

3. **Closure replaces the Virial Theorem.** The relation $\kappa^2 = 2\beta^2$ emerges as the consequence of equitable ledger distribution across the independent degrees of freedom of the directional and omnidirectional carriers.

4. **Precession is geometric, not dynamical.** The secular advance of the pericentre arises as the non-linear algebraic accumulation of internal phase mismatch, bypassing all integration of geodesic equations or expansions in powers of v/c.

5. **Singularities are prohibited.** By enforcing the invariance of the universal transformation limit, the radial distance is bounded ($r \geq R_s/2$). The point $r = 0$ is absent from the admissible relational range.

The traditional separation of spacetime and energy dissolves into a single generative structure. By enforcing epistemic hygiene, RG provides an operational language in which gravitational phenomena can be derived and communicated without surplus ontology.

The hope is that this physical and mathematical clarity returned by RG's methodology will help turn the focus of physical science back to its earliest ambition: to understand the Universe, rather than just to model it.

Final Summary

$$\textbf{SPACETIME} \equiv \textbf{ENERGY}.$$

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